

Mechanistic simulation of heritable expression states, cell division and drug response in single-cell populations

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Abstract

Heritable differences in gene expression decide which cancer cells survive a drug, yet the simulators used to benchmark single-cell methods model neither growth nor division, so cannot generate such states. Here we present Biomodelling.jl 2.0, in which stochastic reaction kinetics run inside growing, dividing, drug-treated cells. Exact, hybrid and adaptive kernels couple to exponential growth, volume-scaled transcription, gene replication, division with binomial partitioning, dose schedules with state-dependent killing, a lineage record, observation models and likelihood-free inference. Expression memory, cell-size scaling and cell-cycle-dependent bursting emerge from this physiology rather than being imposed. One resistance gene with slow promoter switching reproduces the lineage and memory signatures of drug tolerance, and the framework supplies ground truth that benchmarks of memory genes, network inference and perturbation prediction lack. Calibrated to time-lapse measurements of cisplatin-treated cells, it reproduces the dose-invariance of single-cell timing and the kin fate correlations, though six parameters fitted to six fate fractions leave that memory undetermined, and a cycle-gated hazard fits as well. With a built-in optimiser it reproduces the opposite rankings of intermittent dosing reported in a melanoma trial and in xenografts, and shows the ranking to turn on two measurable cell properties: growth under drug and fitness without it.

Isogenic cells differ in their transcriptional states, those differences are often inherited across several divisions, and the inherited state biases what a cell does under stress. Cancer cells that survive targeted or cytotoxic therapy frequently do so without a resistance mutation: they occupy a rare, reversible, drug-tolerant expression state that pre-exists the treatment or is induced by it^{1–5}. Lineage-resolved experiments now measure this memory directly. The fates of sisters and cousins are correlated, clonal barcode diversity is preserved or collapsed depending on whether tolerance pre-exists or is induced, and the timescale of expression fluctuations relative to the cell cycle decides which genes carry memory at all^{6–9}. In parallel, mechanistic analyses of single-cell RNA-sequencing data have shown that transcriptional kinetics can be inferred correctly only when gene copy number, cell age and partitioning at division are modelled explicitly^{10–12}, and evaluations of “virtual cell” models for perturbation prediction keep finding that deep-learning approaches fail to beat simple baselines, largely for want of mechanism and causal structure^{13–16}.

These developments create a demand that existing simulators do not meet. The simulators used to benchmark single-cell methods, among them Splatter, SymSim, scDesign3, SERGIO, dyngen, scMultiSim and GRouNdGAN^{17–23}, model regulation and technical noise but not cell growth, volume-dependent transcription, gene replication or the partitioning of molecules at division; the last of these do place cells on a differentiation trajectory, and lineage simulators such as TedSim and Cassiopeia^{24,25} generate division trees, but the expression attached to those trees is phenomenological and largely Markovian. Expression memory, cell-size scaling, cell-cycle-dependent bursting and persister dynamics therefore have to be imposed on them rather than emerging, and none provides the ground

truth that the methods now being developed for exactly those phenomena require²⁶. Stochastic simulators of growing and dividing cells do exist, and the most complete of them couples reaction kinetics to growth, division and inference from lineage data^{27–30}; what none of them adds is a drug and schedule layer, observation models for the measurements these questions are asked of, and benchmark generation. Agent-based tumour simulators supply the drug response and the spatial structure but not stochastic transcription in dividing cells^{31,32}.

Here we present `Biomodelling.jl` 2.0, a rewrite of a Julia framework whose distinguishing feature, the coupling of stochastic reaction kinetics to exponential growth, volume-scaled transcription and binomial partitioning at division³³, is precisely the mechanism behind heritable expression states. Version 2.0 turns that feature into a complete framework for the mechanistic study of drug tolerance and for single-cell benchmarking. We validate the engine against exact stationary laws and an independent implementation, show that growth, division and partitioning alone reproduce published results on cell-size scaling, partitioning noise, lineage-versus-population statistics and cell-cycle-dependent bursting, and demonstrate that a single resistance gene with slow promoter switching reproduces the quantitative lineage and memory signatures of heritable drug tolerance. We then use the framework as a ground-truth generator for memory-gene detection, network inference, imputation and perturbation prediction; as an inference engine that recovers kinetic parameters from snapshot and lineage data, quantifies the bias of ignoring division, and maps the ridge to which drug parameters are determined by population data; as a model calibrated to published time-lapse measurements and confronted with conditions it was not fitted to; and, with a built-in schedule optimiser, as a testbed in which clinical dosing regimens are compared and better ones are sought.

Results

Layers couple reaction kinetics to growth and division

`Biomodelling.jl` 2.0 is organised in layers (Fig. 1). A model is a list of reactions with mass-action, Hill or custom kinetics and named parameters; promoter states are explicit species, grouped so that they are inherited at division and doubled at replication. Four kernels advance a cell’s state at fixed volume: the exact direct method, fixed-step tau-leaping, a hybrid scheme that falls back to the exact method whenever a leap would produce a negative count, and adaptive tau-leaping with critical-reaction handling. The population layer grows each cell’s volume exponentially, scales propensities with volume so that concentrations are invariant, replicates genes at a set point of the cycle, divides cells by a sizer, adder or timer rule with binomial or over-dispersed partitioning, and controls the population size by Moran-like replacement, free growth or a logistic hazard. A perturbation layer applies dose schedules, state-dependent death, growth inhibition, drug-induced rate changes and genetic perturbations. Every cell’s ancestry, fate and state at birth and division enter a lineage table, from which heritability, autocorrelation, memory timescales, fluctuation-test and clonal-memory statistics follow. Observation models convert true counts into scRNA-seq, smFISH or time-lapse measurements, and the same forward model drives likelihood-free inference. Table 1 sets these capabilities against those of the simulators most used to benchmark single-cell methods.

Table 1. Capabilities of single-cell simulators (✓ native; ○ partial or phenomenological; – absent). SERGIO²⁰, dyngen²¹, scMultiSim²², GRouNdGAN²³, TedSim²⁴, Cassiopeia²⁵, ABM.jl (AgentBasedModeling.jl)²⁸, Biomodelling.jl v1.x³³ and v2.0 (this work). Entries were read from each tool’s publication and documentation, and describe what the tool provides rather than what its formalism could express. Three readings need qualifying. SERGIO is a Python package that returns expression matrices as arrays and scMultiSim is an R package returning SingleCellExperiment-style objects, so neither writes AnnData natively; scMultiSim does carry a differentiation tree, so cells are related by a trajectory even though per-cell division and fate are not simulated. ABM.jl is the closest comparator: it simulates stochastic reaction networks in growing, dividing agents with an adder division rule, binomial partitioning, death and environmental coupling, and its companion work infers kinetic and selection parameters from mother-machine and lineage-tree data²⁹, which the partial marks in an earlier version of this table understated; its ○ entries are for gene replication, which its agent formalism can express but does not provide, and for lineage output, which traces a cell back to its ancestor rather than recording a tree with per-cell fates. Cassiopeia simulates division topologies with heritable fitness and CRISPR barcodes rather than molecular content, and its inference recovers phylogenies rather than kinetic parameters.

Capability	SERGIO	dyngen	scMulti-Sim	GRouNd-GAN	TedSim	Cassio-peia	ABM.jl	v1.x	v2.0
Mechanistic reaction kinetics (SSA / tau-leaping)	○	✓	○	–	–	–	✓	✓	✓
Gene regulatory networks with known ground truth	✓	✓	✓	✓	–	–	✓	✓	✓
Cell growth, volume-scaled transcription	–	–	–	–	–	–	✓	✓	✓
Gene replication / cell- cycle copy number	–	–	–	–	–	–	○	–	✓
Division with molecular partitioning	–	–	–	–	○	○	✓	✓	✓
Heritable promoter states, expression memory	–	–	–	–	○	–	✓	○	✓
Lineage tree with per-cell fates	–	–	○	–	✓	✓	○	–	✓
Drug response: dose schedules, state-dependent death	–	–	–	–	–	–	–	–	✓
Genetic perturbations (knockdown / overexpression)	✓	✓	○	✓	–	–	○	–	✓
scRNA-seq observation model	✓	✓	✓	✓	✓	–	–	✓	✓
smFISH / time-lapse observation	–	–	–	–	–	–	–	–	✓
Parameter inference from data	–	–	–	–	–	○	✓	–	✓
AnnData output	–	✓	–	✓	○	○	–	–	✓

The engine matches exact laws and an independent package

We first verified the kernels against exact stationary laws (Fig. 2). For a birth-death process the simulated distribution of 50,000 cells is indistinguishable from the Poisson law (Fig. 2a); for the telegraph model in an intermediate and in a bursty regime it matches the Beta-Poisson distribution³⁴ (Fig. 2b); and for bursty protein production it matches the negative binomial (Fig. 2c). Across kernels, Kolmogorov-Smirnov distances to the exact law with 20,000 cells lie below the 99% critical value for the direct method, for the adaptive scheme at both tolerances and for the hybrid scheme at the shorter step, and above it for the hybrid scheme at the longer step (0.0160 and 0.0122 against a critical value of 0.0115; Fig. 2d). That is the expected behaviour of an approximate kernel, and it sets the step below which the approximation is not detectable at this sample size; wall-clock costs

are 0.25 s to 0.91 s per 10^4 cells. Simulating the same telegraph model with JumpProcesses.jl gives distributions whose two-sample Kolmogorov-Smirnov statistic against Biomodelling.jl is 0.0034 (99% critical value 0.0163; Fig. 2e), with a 6.2-fold (in favour of JumpProcesses.jl) difference in serial wall-clock time. Those tests exercise one cell at fixed volume. The layer above them, growth, replication, partitioning and the difference between a lineage and a population, has exact solutions of its own for this class of model, which take different closed forms in the two settings^{35–37}. We reproduced four analytically solvable members of that class, with constitutive, bursty and two-state promoter kinetics under a memoryless interdivision time and with volume-scaled synthesis from a replicating gene under a deterministic cycle, in both modes. Along single lineages, where the mother-machine control makes the recorded cells independent and the Kolmogorov-Smirnov test applies as it stands, all 4 samples match the exact law of their mode (distances 0.0023 to 0.0054 against 99% critical values of 0.0081 to 0.0115). A population snapshot is not an independent sample, because its cells share ancestors, so the independent critical value understates the true one and the population mode is judged on the agreement of its replicate means with the exact mean. Over the four models those means sit within 2.5 standard errors of the exact values (18 of the 20 individual samples also fall below the independent critical value, the largest distance being 0.0170). Scored against the exact law of the other mode, the same samples give distances at least 6 times larger, so the comparison has ample power to tell the two settings apart. The two differ because a snapshot of a growing population over-weights cells that have just divided and so just lost half their molecules: the mean molecule number is constitutive production 13.33 against 10.00; bursts of four molecules 13.33 against 10.00; two-state promoter 8.89 against 6.67; volume-scaled synthesis with gene replication 25.64 against 23.96 along a lineage and in a population respectively (Supplementary Note 14 and Supplementary Fig. 6). Population simulations scale linearly with the number of cells: 2.6 s for 5000 cells $\times$ 10 telegraph genes $\times$ 200 steps (hybrid kernel) with four threads (Fig. 2f and Supplementary Table 3). Scaling in the number of reactions depends on the kernel and is close to linear only for the hybrid one (0.58 s to 6.4 s for the hybrid kernel, a factor of 11 for the same factor of 10 in genes; 1.04 s to 31.4 s for the direct kernel, a factor of 30 for the same factor of 10 in genes), because the direct method draws one event at a time while the hybrid kernel leaps over the non-critical channels in a fixed number of operations.

Physiology generates memory, scaling and cycle effects

The physiology of the population layer suffices to produce properties of single-cell data that phenomenological simulators must impose by hand (Fig. 3). In a single lineage of a slowly switching gene, mRNA and protein track volume within a cycle and are halved at division, while the promoter state, inherited by both daughters, persists over many divisions (Fig. 3a). For a fast-switching gene without replication, transcript numbers scale linearly with cell volume, so that concentration is homeostatic; adding mid-cycle gene replication doubles transcription in the second half of the cycle and lifts counts of large cells above the volume-scaling line (Fig. 3b), the copy-number effect that mechanistic fits to scRNA-seq data must account for¹⁰. Correlations of protein concentration at division between mothers and daughters, sisters and cousins fall from 0.94 to 0.02 as the promoter switching rate crosses the inverse cell-cycle time (Fig. 3c), mother-daughter and sister pairs correlating similarly at slow switching and mother-daughter pairs more strongly as switching quickens, cousins least, and mRNA less heritable than protein. The memory timescale extracted from the generation-resolved lineage autocorrelation is set by the promoter switching time and is nearly independent of the cell-cycle time as long as memory outlasts the cycle; when the two are comparable the estimate is bounded below by one cell cycle, the resolution of the autocorrelation (Fig. 3d). As reported for real lineages¹¹, the coefficient of variation of a stable protein’s concentration is larger across the population than along single lineages, whether or not cell-cycle times are noisy, and over-dispersed partitioning adds variability in both perspectives (Fig. 3e). Telegraph parameters fitted separately to cells before and after replication move in the direction gene replication implies, but not by the factor it alone would give (Fig. 3f).

Between the one-copy and the two-copy subsets the fitted transcription rate rises 2.0-fold, of which 1.4-fold is the larger volume of cells that have already replicated, leaving 1.4-fold per unit volume rather than the twofold that doubling the copy number implies; the fitted activation rate rises 1.4-fold over the same comparison, and the pooled fit that ignores the cell cycle returns a transcription rate above both subsets. Part of that shortfall is misspecification rather than copy number, and the figure shows it: two independently switching alleles are not one telegraph gene, and neither subset fit recovers the truth (0.5, 1.5 and 2 transcripts per unit volume per allele, against 0.59, 1.89 and 2.4 fitted to the one-copy cells and 0.65, 3.27 and 4.8 pooled). What the panel demonstrates is that a cell-cycle-blind fit is biased, not how large the copy-number effect is.

One resistance gene reproduces tolerance signatures

To test whether heritable expression states suffice to reproduce the lineage and memory signatures of drug-tolerant persistence, we coupled a resistance gene with slow promoter switching (memory timescale 45 time units, cell-cycle time 20) to a death hazard that rises with dose and is suppressed by the concentration of the gene's protein product (Methods). Under continuous, pulsed and reduced-dose schedules the population first collapses as low-expressing cells die, then regrows from high-expressing survivors (Fig. 4a); kill curves are biphasic with a surviving fraction that falls with dose (Fig. 4b). Between doses 0.5 and 2 the net rate of population decline over the first 30 time units of treatment rises from 0.023 to 0.050 per time unit (2.1-fold), while the mean division time of cells born under drug changes by 0.0% and the mean time to death of killed cells moves only from 8.5 to 6.2 time units (Fig. 4c; the single-cell timing distributions are in Supplementary Fig. 1). This reproduces the observation that population decay depends strongly on dose even though single-cell division and death times barely do⁶, because fate is biased by states that exist before the drug arrives. Unlike the growth-arrested persisters of the canonical definition¹, the survivors here divide at an essentially unchanged rate (Fig. 4c, inset), so the model reproduces the heritable-state selection signatures of tolerance rather than its proliferative-arrest hallmark.

Related cells accordingly share the fate of their lineages. Among cells born in the 25 time units before the drug, one whose sister founded a surviving lineage does so itself with probability 0.39, against a marginal probability of 0.08 (4.8-fold enrichment over 1830 sister pairs; 2.9-fold for cousins), whereas in a fast-switching control, in which survival is rarer, the conditional and marginal probabilities coincide (0.03 versus 0.02; Fig. 4d). When tolerance pre-exists, survivors descend from a minority of founder clones and the effective number of clones drops from 491 to 89; when tolerance is drug-induced, so that the drug raises the promoter activation rate of every cell, barcode diversity is largely preserved (165 of 492; Fig. 4e). This is the signature that distinguishes the two origins in barcoding experiments^{2,6}; the follow-up method that traces resistant fates back to their drug-naïve precursors puts those precursors at an initial frequency of about 1 in 1,000 to 1 in 10,000 cells³⁸, the regime the pre-resistant fraction of the melanoma case study below is set to.

Disrupting expression memory changes the outcome, and when it is disrupted matters. Accelerating promoter switching 20-fold from two cell-cycle times before the drug does not reduce the number of surviving clones if the acceleration stops when the drug arrives: it raises them, from 98 to 120 (mean of 5 seeds each, $P = 0.009$, Welch's t test). Reshuffling spreads the resistant state over more lineages and moves the resistant fraction towards its stationary value (0.09 to 0.11), so more clones carry a protected cell when the drug arrives, even though protection itself is untouched. Maintaining it during exposure reduces the count to 7 clones (10 cells, against 792; Fig. 4f), because resistant cells now leave the protected state within a fraction of the time they need to found a colony. In the model, memory must be disrupted while the drug is present.

Two published experiments bear on this prediction and they point in opposite directions, so it is worth being explicit about which perturbation each one performs. A CRISPR screen for modulators of single-cell variability

found that inhibiting DOT1L *before* adding a BRAF inhibitor produced more therapy resistance than giving the two together³⁹, which is the ordering the model predicts. A lineage-tracing study that identified PI3K and TGF- β signalling as switching modulators found the opposite for its own pretreatment: a PI3K inhibitor applied before targeted therapy left less resistance than targeted therapy alone⁹. The two results are not in conflict with each other, and only the first is a test of what is simulated here. The perturbation in the model is symmetric: it multiplies the activation and the inactivation rate by the same factor, so it accelerates relaxation towards the stationary resistant fraction without moving that fraction, and its only effect before the drug is to spread the resistant state over more lineages. A modulator that shifts the balance between the states is a different perturbation, and a pretreatment that drives cells out of the primed state lowers the number of protected cells at the moment the drug arrives, which reduces resistance whether or not it is maintained. The prediction here is therefore specific to perturbations that speed switching without biasing it, of which DOT1L inhibition appears to be one and PI3K inhibition is not; distinguishing the two experimentally requires measuring the resistant fraction as well as the switching rate, which memory-sequencing and lineage-tracing assays do. Compartment theory for phenotypic switching already predicts that an agent which perturbs the retention of the resistant state can turn treatment failure into success⁴⁰; what the simulations add is the timing requirement and a lineage-resolved readout, the number of surviving clones, by which it can be tested. A transient pulse enriches high-expressing cells and, with slow switching, the enrichment persists for many generations after the drug is removed, whereas with fast switching it relaxes within a few cycles (Fig. 4g): phenotypic selection with stable inheritance, as in the original MGMT model⁴¹. Finally, scanning release periods and doses of intermittent schedules (Fig. 4h), the most effective schedule is release period 0, dose 2 (growth rate +0.0038 per time unit, ± 0.0007 over 5 seeds) for pre-existing tolerance, release period 0, dose 2 (growth rate -0.0151 per time unit, ± 0.0012 over 5 seeds) when the resistant state carries a fitness cost, and release period 0, dose 1 (growth rate +0.0089 per time unit, ± 0.0001 over 5 seeds) for drug-induced tolerance. Each point of the scan is the mean of 5 independent seeds. The gap between the best schedule and the next best is 0.0003 to 0.0006 per time unit, against a standard error of 0.0001 to 0.0008 on the difference, so the ranking of the top two schedules is resolved in 1 of the 3 mechanisms; elsewhere the scan identifies a region of good schedules rather than a single best one. Without a fitness cost nothing beats continuous dosing at the highest dose, since every release period lets the resistant subpopulation regrow (the drug does not act on the switching rates in this model; when it also stabilises the resistant state, as in the melanoma case study below, holidays regain a small advantage even without a cost). With a fitness cost the population declines under every schedule at dose ≥ 0.5 , and release periods of up to 10 time units match continuous dosing (-0.0151 per time unit under continuous dose 2), because cells that leave the resistant state during a holiday are killed on re-exposure. For drug-induced tolerance the ranking of doses reverses: the highest dose induces tolerance in more cells than it kills, so dose 1 outperforms dose 2 at every release period (release period 0, dose 1 (growth rate +0.0089 per time unit, ± 0.0001 over 5 seeds), against +0.0123 for continuous dose 2), the intermediate-dose optimum reported previously⁴²; the benefit of release periods reported in that work does not appear here, because in this model the induced state decays at the same rate with and without drug, so a holiday only postpones the death of reverting cells while all cells grow.

The hazard above does not depend on where a cell is in its division cycle, and the calibration below will show that fate data cannot exclude one that does, so we repeated these simulations with three quarters of the hazard confined to a window around mid-cycle. Population decay still rises with dose (-0.001 to +0.023 per time unit over doses 0.5 to 2, against +0.023 to +0.050 without the gate) and the mean time to death stays nearly dose-invariant (8.4 to 9.9 against 6.2 to 8.5 time units), so neither signature depends on a cycle-blind hazard. Sisters share a birth time and so a cycle phase, which lets a gated hazard correlate their fates with no inherited state; it does not do so to any useful degree, lifting the sister concordance of the memoryless fast-switching control only to

+0.008 above independence (from +0.000), against +0.054 for the memory gene, so the kin signature still reads expression memory rather than the cycle. The schedule conclusions are the fragile ones, because the best dose is unchanged in 3 of 3 models, but the best release period is not. With a fitness cost of resistance and with drug-induced tolerance, the holidays that won without the gate collapse to continuous dosing, because a gated hazard kills the reverting cells more slowly on re-exposure (Supplementary Note 13 and Supplementary Fig. 5).

Simulated ground truth benchmarks inference methods

Because every cell's clone, lineage and true state are known, the framework supplies ground truth that observational data cannot (Fig. 5). For 40 independent genes whose memory times span three decades, MemorySeq-style clonal variance scores computed on true concentrations rise from 1.0 for genes with memory far below the cell-cycle time to a mean of 29.6 for genes with memory beyond a hundred cell-cycle times, the highest single gene reaching 44.5 (38.4 on sequenced counts), and the ranking survives computation on sequenced counts at 20% capture efficiency (Fig. 5a); the score separates genes with memory longer than the cycle from the rest with an area under the receiver operating characteristic (ROC) curve of 1.00 on true concentrations and 0.99 on sequenced counts. For a 30-gene network, small enough that the methods were run with their default settings and the ranking of datasets rather than the absolute accuracy is what should be read, simulated at fixed volume, in a dividing population and after sequencing, correlation-based and GENIE3 inference lose most of their accuracy in the dividing population when raw counts are used (area under the precision-recall curve 0.52 to 0.14 for Pearson correlation and 0.32 to 0.13 for GENIE3, against 0.10 for a random ranking of undirected pairs and 0.05 for directed edges), because volume and gene copy number induce positive correlations between all genes. Normalising counts to concentrations removes the volume effect but recovers little (0.18 and 0.18), since the dependence on copy number and cell age remains; regressing expression on the true cell-cycle covariates, which the simulation provides and which cell-cycle scoring approximates in real data, recovers most of the loss (0.46 and 0.30). Sequencing at 15% capture efficiency lowers accuracy further (0.11 raw and 0.24 library-size-normalised for Pearson correlation; 0.07 and 0.08 for GENIE3; Fig. 5b). One comparison in that panel runs the wrong way: for Pearson correlation the library-size-normalised sequenced counts score above the true concentrations (0.24 against 0.18). Dividing counts by the library size removes cell-to-cell scale, and in a dividing population that scale is dominated by volume and gene copy number, so the normalisation acts as a coarse correction for the growth confound that dividing by the true volume does not make. It is not evidence that sequencing helps; the fixed-volume ceiling (0.52) is above both, and the cell-cycle regression (0.46) is above both as well. Imputing the sequenced counts before inference recovers part of that loss: MAGIC raises the area under the precision-recall curve from 0.11 to 0.32 for Pearson correlation and from 0.07 to 0.18 for GENIE3, and kNN-smoothing gives 0.16 and 0.12 (Fig. 5c). Neither reaches the accuracy obtained at fixed volume, nor that of regressing out the cell-cycle covariates, so imputation mitigates the sequencing loss without touching the growth confound that produces most of it³³. Knocking down the five most connected regulators of a 20-gene network, again small and run at default settings, gives true fold changes that a no-change baseline predicts with a mean coefficient of determination of -0.16. Propagating observational correlations beats that baseline for 4 of 5 knockdowns, with coefficients of determination from 0.09 to 0.43, and fails on the remaining one (-2.39), so its mean over the five is -0.23 against a median of 0.30 (Fig. 5d). A predictor that is useful for most perturbations and badly wrong for one is precisely what an out-of-distribution evaluation has to expose, and exposing it needs the causal ground truth that the simulation supplies^{13,14}.

Inference recovers parameters and exposes division bias

On a snapshot of 2,000 non-dividing cells, ABC-SMC with log-uniform priors over three decades and the Wasserstein distance between simulated and observed counts⁴³ concentrates on the true telegraph parameters and

266 agrees with the exact maximum-likelihood estimate (posterior medians (0.31, 0.60, 20.9) for truth 0.3, 0.6, 20; Fig.
 267 6a), the broadest posterior falling on the promoter inactivation rate, which a single snapshot constrains least.
 268 On a snapshot of a growing and dividing population with volume-scaled transcription and mid-cycle replication,
 269 the naive telegraph fit is biased ($k_{\text{on}} = 0.47$, $k_{\text{off}} = 1.95$, $k_{\text{tx}} = 71.5$) because volume and copy number inflate
 270 counts and their dispersion, whereas ABC with the population simulator as forward model recovers the truth
 271 ((0.30, 0.68, 20.8); Fig. 6b). Drug parameters follow from population data in the same way: from a kill curve
 272 and the concordance of sister fates, the posterior over the maximal death rate and the protection threshold is
 273 a ridge (weighted correlation of the two log parameters -0.85) that contains the truth (Fig. 6c), and posterior
 274 predictive kill curves simulated at the posterior median reproduce the observed one (Fig. 6d). The two summaries
 275 constrain the product of death rate and unprotected fraction, so separating the parameters requires measuring
 276 the protective protein: an experimental-design question the forward model answers before an experiment is run
 277 (the tolerance and acceptance-rate schedules of all three ABC runs are in Supplementary Fig. 2).

278 **Calibration to time-lapse data is tested on held-out data**

279 A simulator is useful for treatment questions only if it can be fitted to the data a laboratory produces and then
 280 predict what it was not fitted to. We therefore calibrated the resistance-gene model to time-lapse measurements
 281 in which U2OS cells were tracked for three days under 7, 10 and 13 μM cisplatin and every cell present at drug
 282 addition was scored as dying, dividing or surviving without dividing⁶ (Supplementary Table 4). Six parameters,
 283 the two promoter switching rates, the maximal death rate, the EC_{50} and Hill coefficient of death and the IC_{50}
 284 of growth arrest, were fitted to the fate fractions at 7 and 13 μM by simulated moments with common random
 285 numbers, under a weak prior keeping the memory of the resistant state near the two to three generations that the
 286 same study inferred from lineage correlations (Methods). The calibrated model reproduces the training fractions
 287 with a root-mean-square error of 0.077 (memory 2.8 generations, resistant fraction 15.2%, EC_{50} 16.7 μM ; Fig.
 288 7a,f). Three related numbers appear in this Article and are not interchangeable. Every root-mean-square error
 289 quoted in the text, here and at the held-out concentration, is the residual of the seed-averaged fate fractions
 290 pooled over every fraction of the concentrations named, four seeds per condition. The colour scale of Fig. 7f is
 291 the quantity the optimiser minimised, which is evaluated on a single seed and adds the memory prior: at the
 292 optimum it is 0.067, made up of 0.061 of single-seed training error and 0.006 of prior penalty. The comparison
 293 of the two killing mechanisms in Supplementary Note 12 uses that single-seed training error for both models,
 294 so that they are scored by the same rule as the fit itself. That error should be read against the precision of the
 295 measurement: the observed fractions carry a mean binomial standard error of 0.028 and repeated simulations
 296 differ by 0.015, so the fit is about 2.4 times the combined noise and the model does not reproduce the training
 297 data within error. At the held-out concentration the root-mean-square error is 0.093, and the discrepancy is
 298 specific rather than diffuse: the model matches the fraction surviving without dividing but over-predicts death
 299 and under-predicts division (observed died, divided and survived 0.28, 0.31 and 0.41; predicted 0.39, 0.20 and
 300 0.41; Fig. 7a), so it captures the ordering of the three concentrations while misplacing the split between death
 301 and division at the intermediate one.

302 Six parameters fitted to six numbers are not all determined by them. Moving one parameter at a time from
 303 the fitted values, with the others held there, the fate fractions constrain the death parameters to within about
 304 a factor of 2 but leave the two switching rates free over roughly a factor of 6 (Supplementary Fig. 3a), which
 305 is why the memory of the resistant state was given a prior rather than left to the fate data. Holding the death
 306 parameters at their fitted values, memories from 1.0 to 9.0 generations paired with resistant fractions from 1%
 307 to 25% all describe the fate fractions within the noise of the measurement (Supplementary Fig. 3b), because a
 308 shorter memory with more resistant cells and a longer memory with fewer are hard to tell apart from fates alone.

309 The parameters quoted above should therefore be read as one member of a family of fits that describe these fate
310 fractions equally well, and the biological reading of the memory rests on the lineage correlations that set the prior,
311 not on the fate fractions. Halving the integration step changes the simulated fractions by 0.005, less than the
312 seed-to-seed spread, so the discretisation is not a factor (Supplementary Fig. 3c).

313 A second mechanism can produce the same fates. Cisplatin kills preferentially during replication, so where a cell
314 sits in its division cycle when the drug arrives is a competing explanation both for the dose-invariance of death
315 timing and for the cells that survive three days without dividing: a hazard confined to part of the cycle spares
316 whatever never enters it, with no heritable state at all. We therefore refitted the model with a cell-cycle-dependent
317 hazard, adding one parameter, the fraction of the hazard that is independent of cycle position, with killing
318 otherwise peaking at mid-cycle, under the same objective, bounds, budget and seeds (Methods). The two fits
319 describe the training fractions equally well (0.064 against 0.061, against a mean sampling error of 0.028 in the
320 measurement), and the cycle-dependent fit does not improve the held-out prediction either (0.106 against 0.095).
321 The fitted cycle-independent fraction is 0.34, but values from 0.20 to 0.80 all sit within one sampling error of the
322 best fit, so the fate fractions do not determine it (Supplementary Note 12 and Supplementary Fig. 4). Three fate
323 counts per concentration cannot separate inherited protection from cell-cycle position; the measurement that can
324 is the cycle position of each cell at the moment the drug arrives, which a cycle reporter records directly.

325 Properties that were not fitted can then be compared with the data. The predicted kill curves (Fig. 7b) are
326 1.5-fold steeper at 13 μ M in their first two days than in their second half, but they do not plateau, as HCT116
327 cells do after about 100 h. Death fates are correlated between sisters (0.09), first cousins (0.06) and second cousins
328 (0.05), and the correlation decays to 0.02 for third cousins (95% confidence interval +0.000 to +0.043 over 3
329 seeds, a range that contains zero) and 0.00 for unrelated cells (Fig. 7c). That is the pattern reported for HCT116
330 lineages, in which fate correlations are present for first and second cousins and absent for third cousins, with
331 memory inherited over at least two to three generations⁶; the comparison is qualitative on both sides, because the
332 source reports these correlations graphically and gives no numeric coefficients or confidence intervals to match.
333 One caveat belongs beside this agreement. Cells here draw their cycle times independently, so no cycle timing is
334 inherited and every correlation between related cells is carried by the inherited promoter state. Cell-cycle duration
335 is in fact heritable over several generations, and its inheritance alone produces strong cousin and second-cousin
336 correlations even when mother-daughter correlations are weak^{44–46}. Coupled to a cycle-gated hazard, that is a
337 complete alternative explanation for the correlations fitted here, and the model as built cannot produce it. The
338 mean time to death changes by 8% between 7 and 13 μ M while the death fraction rises from 0.13 to 0.59 (Fig. 7d),
339 the dose-invariance of single-cell timing that motivated the original study, and it arises here because dose changes
340 the fraction of unprotected cells rather than their kinetics. The plateau the kill curves miss is the clearest failure
341 of the minimal model: reproducing it requires cells to stay in the resistant state for longer than the memory of 2.8
342 generations allows, and that memory was set by the prior taken from the lineage correlations rather than by the
343 kill curves. The two observations therefore pull in opposite directions within this minimal model: correlations
344 that vanish by the third cousin call for a short memory, and a kill curve that flattens calls for a long one. A single
345 switching gene cannot satisfy both, which is a concrete and testable limitation, and points to a drug-induced
346 stabilisation of the resistant state, for which there is independent evidence², as the mechanism a lineage-tracing
347 experiment should look for. The calibration landscape (Fig. 7e) shows how the fitted death rate and EC_{50} are
348 constrained by the fate fractions alone; the protocol applies to any observable a laboratory records.

Schedule optimisation is consistent with clinical outcomes

With a calibrated model the framework becomes a schedule testbed. We first asked which schedule best delays regrowth of a melanoma-like population with a rare, heritable pre-resistant state² treated at a fractional-killing dose, comparing continuous dosing, the intermittent schedule of the SWOG S1320 trial, in which patients received a continuous 8-week lead-in and were then randomised to continuous dosing or to cycles of 3 weeks off and 5 weeks on⁴⁷, and adaptive therapy, in which the drug is withdrawn when the population has halved and resumed when it regrows to its initial size⁴⁸. Because adaptive therapy deliberately lets the tumour regrow, schedules are compared by the time after randomisation at which the population exceeds 120% of its pre-treatment size (loss of control), with the RECIST-like time to progression from the nadir reported alongside (Methods).

The answer depends on two properties of the resistant cells (Fig. 8a–d). When the resistant state is fully protective and carries no fitness cost, as lineage data indicate for pre-existing tolerance⁶, resistant cells grow at the same rate with or without drug, so a holiday cannot cost the resistant subpopulation anything and can only let sensitive cells regrow to be killed again. Even so, both interrupted schedules keep control longer than continuous dosing: the three lose control after 23, 26 and 27 weeks, with the intermittent schedule delivering 65% of the cumulative dose of continuous dosing over the full 60-week simulation. The reason is a second drug effect rather than the fitness of resistant cells: the drug stabilises the resistant state tenfold by suppressing the promoter inactivation rate (Methods), so continuous exposure keeps cells resistant that a holiday allows to revert. That effect is set in the model, not fitted, and it is what makes holidays worth anything at all in this arm. When resistant cells pay a 50% fitness cost in the absence of drug, a coarse stand-in for the drug dependence that made intermittent dosing superior in vemurafenib-treated patient-derived xenografts, where withdrawal drove established resistant tumours to regress rather than merely grow more slowly⁴⁹, holidays let sensitive cells outcompete them: none of the three schedules loses control within the 52-week follow-up (52+, 52+ and 52+ weeks, censored), and adaptive therapy achieves this with 36% of the cumulative dose of continuous dosing. When resistant cells are only partly protected and grow at half speed under drug, holidays release them: continuous dosing keeps control for 52+ weeks against 49 for the intermittent schedule and 43 for adaptive therapy, the ranking observed in the trial, where continuous dosing gave a median progression-free survival of 9.0 months against 5.5 months for intermittent dosing. Gating three quarters of the death hazard to a window of the cell cycle, as in the persister simulations above, leaves these comparisons standing, whether or not the maximal hazard is raised to hold the cycle-averaged hazard fixed (Supplementary Note 13 and Supplementary Fig. 5): the best schedule is unchanged in 3 of 3 mechanisms and the model still favours continuous dosing under partial protection (52 against 43 weeks, from 52 and 49). The memory of the resistant state, five net population doublings, is an assumption rather than a measurement, and it is what decides whether resistant cells revert during a three-week holiday; scanning it from two to twelve doublings leaves the ranking unchanged and narrows the advantage of holidays from 9 weeks at 2 net doublings of memory to 1.7 weeks at 12 (Supplementary Note 15). That the benefit of holidays should depend on a cost of resistance is established by compartment models of adaptive therapy, which also show that tumour turnover modulates how large that cost must be⁵⁰. Here the dependence emerges from single cells whose resistance is a heritable expression state rather than a fixed compartment, the cost and the degree of protection are properties of cells that lineage experiments measure, and the third case shows that partial protection reverses the ranking even when a cost is absent. The optimiser makes the dependence explicit (Fig. 8e): over intermittent schedules of any period and duty cycle after the lead-in, the best schedule is a period of 2.2 weeks with 39% of the time on drug (27 weeks to loss of control) without a fitness cost, a period of 1.0 weeks with 47% of the time on drug (52 weeks to loss of control) with a fitness cost and a period of 1.0 weeks with 100% of the time on drug (52 weeks to loss of control) with partial protection. A bacterial-persistence model adapted to colorectal-cancer lines reaches the same conclusion by another route: an intermediate dose with release periods is optimal, because the rate of transition

into the persistent state rises with concentration and then saturates⁴². The agreement is partial and the difference is informative. Where tolerance is drug-induced, the optimiser here also selects an intermediate dose over the highest one, for the same reason. Release periods, however, help there only when the drug changes the kinetics of the resistant state, either by stabilising it, as in the melanoma arm above, or by inducing it at a rate that saturates, as in that model; with a state whose decay rate is the same with and without drug, a holiday merely postpones the death of reverting cells while every cell grows, and the optimiser rejects it. The model therefore reproduces both the clinical and the preclinical outcome and shows their difference to depend on measurable properties of the resistant cells, their growth under drug and their fitness without it, both properties of cells rather than of the tumour, which growth and lineage experiments can bound, within the identifiability limits of Fig. 6c, before a schedule is chosen. The mechanism is set in each case, not inferred from the outcome, so this is a demonstration that the ranking follows from cell properties rather than a claim that these particular cells carry them; and because the trial and the xenograft also differ in drug and species (dabrafenib plus trametinib in patients, vemurafenib monotherapy in mice), their contrast is not attributable to resistant-cell fitness alone.

The second case study concerns glioblastoma, where the RTOG 0525 trial compared the standard temozolomide regimen (150–200 mg/m² on days 1–5 of 28-day cycles) with a dose-dense regimen (75–100 mg/m² on days 1–21) that delivers about twice the cumulative dose and was designed to deplete MGMT, and found no difference in survival for either MGMT-methylated or unmethylated tumours⁵¹. We simulated both regimens with one-compartment pharmacokinetics (elimination half-life 2.1 h in the cerebrospinal-fluid population model used here⁵²; the plasma value is 1.8 h⁵³) on populations in which 1% (methylated) or 30% (unmethylated) of cells express MGMT. Those two fractions are illustrative values that bracket low and high expression, not measurements, and the idealisation matters: promoter methylation shifts MGMT expression as a graded quantity across cells rather than switching a discrete subpopulation on, so the whole dose-dense prediction below rests on treating it as two states. With that caveat, with MGMT protein protecting cells from the drug⁴¹ and either stable or consumed by it stoichiometrically, one molecule per lesion repaired, as a suicide enzyme is, so that the pool follows the cumulative exposure rather than the peak concentration (Fig. 8f,g). Lesions are not tracked as a state variable: they are taken to form in proportion to dose and to be repaired at once, so “one molecule per lesion” enters only through the rate of consumption, and no unrepaired damage accumulates. The rate constant of that consumption is not fitted to data; it is set so that consumption at the standard bolus peak matches the first-order parameterisation it replaces, which makes the two mechanisms comparable at the same initial rate and isolates how each integrates over time. With stable MGMT, the depth of response over six cycles is 0.20 log₁₀ units for the standard regimen and 1.00 for the dose-dense regimen in the methylated population (0.08 and 0.10 in the unmethylated one), and a 21-day regimen at the standard cumulative dose achieves no measurable kill (0.00 log₁₀ units): the death hazard rises steeply with concentration, so spreading the same dose over more days loses efficacy, and the advantage of the dose-dense arm in the model is entirely its larger cumulative dose. With stable MGMT the two arms therefore differ little in the unmethylated population (0.08 against 0.10 log₁₀ units), as in the trial. Consumption changes that. Because one molecule is spent per lesion repaired, the pool follows the cumulative exposure, which the dose-dense regimen more than doubles: in the methylated population its response deepens to 1.52 log₁₀ units against 0.22 for the standard regimen, and in the unmethylated population, where 30% of cells start protected, the same depletion turns a regimen that barely worked into one that controls the population (1.08 against 0.18 log₁₀ units; final size 0.1 against 1.1 times the initial size, and 4.9 and 3.6 with stable MGMT). Modelled as the suicide enzyme it is, MGMT therefore predicts a clear dose-dense advantage in both populations, which RTOG 0525 did not find. Read as a bound rather than as a fit, that disagreement is informative: either temozolomide consumes MGMT in tumours far more slowly than this stoichiometry allows, or the benefit of depleting it is offset by something the model leaves out. At the standard cumulative dose the number of dosing

days per cycle that minimises the final population is 9 (methylated) and 9 (unmethylated) with stable MGMT, and 9 and 5 when MGMT is consumed (Fig. 8h). These simulations are not fits to the trial, whose end points depend on much that the model omits, dose-limiting toxicity and the dose reductions it forces among it. What they add is a sharper version of the rationale for dose-dense dosing: the benefit it is supposed to deliver exists in the model only at consumption rates that the trial's null result appears to exclude. The quantity the argument rests on is the per-lesion molecular rate constant of that consumption, which has not been measured directly, although bulk depletion time-courses have: protracted temozolomide lowers alkyltransferase activity in peripheral blood mononuclear cells by about 63% at 14 days and 73% at 21 days⁵⁴, a compressed 1,000 mg/m² schedule takes it to 18% of initial levels while O⁶-methylguanine accumulates in both blood cells and tumour biopsies⁵⁵, and in orthotopic GBM43 xenografts tumour MGMT activity is depleted by day 6 on both the standard and the 21-day schedule and stays suppressed only on the 21-day one⁵⁶. The last of these is the sharpest comparison available, and it makes the disagreement more specific than the trial alone does: the deeper and longer tumour depletion that the dose-dense schedule is supposed to deliver was confirmed in that model, and survival still did not improve. Either the depletion measured in bulk is not the per-cell depletion that matters for killing, or the benefit of depleting MGMT is offset by something the model leaves out.

Discussion

Biomodelling.jl 2.0 fills a gap in the single-cell toolkit: a simulator in which expression memory, cell-size effects, cell-cycle dependence and drug-tolerant persistence follow from explicit biophysics instead of being assumptions of a generative model. It is worth separating what is new here from what it confirms. No individual ingredient is new, and the claim is not that any of them is: stochastic kinetics in growing and dividing cells have been simulated before^{27,28,33}, lineage-tracing data have their own simulators^{24,25}, compartment models of adaptive therapy established that the benefit of treatment holidays depends on a cost of resistance and on turnover^{48,50}, and phenotypic-switching theory established that perturbing the retention of a resistant state can decide a treatment outcome⁴⁰. What is new is that this is the first package to combine these mechanisms with joint inference and schedule optimisation in a single forward model. Version 1 of this framework already simulated stochastic gene networks in growing and dividing cells and generated synthetic scRNA-seq for imputation and network-inference benchmarking³³; what version 2 adds to it is gene replication and cell-cycle copy number, division by sizer, adder or timer rules rather than a timer alone, inherited promoter states, the lineage table and its statistics, the drug and pharmacokinetic layer with state-dependent killing, likelihood-free inference, and the schedule optimiser. Relative to the closest external comparator, an agent-based framework that already couples stochastic kinetics to growth, adder division, partitioning and inference from lineage data^{28,29}, what is added is the drug and schedule layer, the observation models, gene replication, and the benchmark-generation use. Two consequences follow from the combination rather than from any part of it. It makes a resistance phenotype a heritable expression state with measurable kinetics instead of a compartment with an assumed rate, and it supplies ground truth of a kind observational data cannot: the same simulation gives known memory genes, known networks, known perturbation effects and the true lineage tree, which is what the quantification of how growth and sequencing destroy network inference, and of how far cell-cycle regression recovers it, depends on. It also makes the calibrate-then-predict protocol available for schedule questions, which turns a mechanism into a falsifiable prediction: here it identifies the growth of resistant cells without drug and their residual growth under drug as the two properties that decide whether holidays help, and points to the rate at which temozolomide consumes MGMT as the measurement that would separate the two glioblastoma regimens. That difference has practical consequences. As a hypothesis engine for non-genetic drug tolerance, the framework answers which schedule, which

479 combination or which memory-modulating pretreatment minimises the long-term fitness of a treated population,
480 with cell-level rules that can be matched to time-lapse and lineage-tracing data. As a ground-truth generator, it
481 produces data with known memory genes, known networks, known perturbation effects and known lineage trees,
482 which is what MemorySeq-style scores, lineage-aware models and virtual-cell benchmarks need in order to be
483 evaluated fairly, and what existing simulators cannot supply. As a likelihood-free inference engine, it estimates
484 mechanistic parameters from data that are now routinely collected, while quantifying the bias of simpler models.

485 The calibration and schedule studies set out a workflow for in silico single-cell pharmacology: declare the training
486 and validation observables before fitting, fit by simulated moments or approximate Bayesian computation with
487 common random numbers, validate on conditions and statistics that were not fitted, and only then compare
488 regimens, reporting outcomes that map onto clinical end points. The case studies show why mechanism matters.
489 The same question, continuous or intermittent dosing, has opposite answers depending on two properties of the
490 resistant cells, their growth under drug and their fitness without it, and a trial result and a xenograft result
491 that appeared to contradict each other are both reproduced once those properties are set. They are set in
492 each case rather than inferred from the outcomes, so the case studies show that the ranking follows from cell
493 properties, not that any particular tumour has them. Neither anchor is as one-sided as a single end point makes
494 it look. In SWOG S1320 the progression-free survival advantage of continuous dosing did not carry through to
495 survival after progression, which favoured the intermittent arm⁴⁷; and in the xenografts the resistant tumours
496 were drug-addicted, regressing on withdrawal rather than merely growing more slowly⁴⁹, which is the extreme of
497 the fitness cost that the second mechanism here represents. Because the framework simulates the lineage-resolved
498 data from which such properties are measured, it can also say which experiment would settle the question before
499 a trial is designed.

500 Several limitations delimit this release. Cells are well-mixed compartments without spatial structure or cell-cell
501 signalling; the pharmacology is a one-compartment model; growth is exponential with phenomenological size
502 control; and the inference layer, although general, relies on user-chosen summary statistics, so posterior calibration
503 should be checked with predictive simulations. The persister case study is deliberately minimal, a single resistance
504 gene with slow switching, and its calibration used fate fractions transcribed from a published study rather than
505 raw tracks. Its death hazard was first written without cell-cycle dependence, which matters for a drug such as
506 cisplatin whose lethality is enhanced during replication. Refitting with a cycle-dependent hazard shows that the
507 two mechanisms are not separable by these data: a model in which killing is confined to the replication window
508 accounts for the fate fractions as well as heritable protective expression does, and the fraction of the hazard that is
509 cycle-independent is not determined by them at all. Cell-cycle position at the moment the drug arrives is therefore
510 a live alternative to inherited protection, and the experiment that distinguishes them is a cycle reporter read
511 at drug addition rather than more fate counts. Cells draw their interdivision times independently, so cell-cycle
512 duration is not inherited, and every fate correlation between related cells in these simulations is carried by the
513 inherited promoter state. That is a real restriction and not only a simplification. Cell-cycle duration is heritable
514 over several generations in mammalian cells, and its inheritance alone produces strong correlations between
515 cousins and second cousins even where mother-daughter correlations are weak^{44–46}; combined with a cycle-gated
516 hazard it is a complete competing explanation for the second-cousin fate correlations that the calibration in
517 Fig. 7 reproduces, and the model as built cannot generate it, so the fit has no choice but to load all of that
518 correlation onto expression memory. Chromatin states or cycle timing with their own inheritance rules would
519 let the two be separated, and are the extension this points to. Six parameters fitted to six fractions leave the
520 switching rates weakly determined, so the memory of the resistant state rests on the lineage correlations used as a
521 prior rather than on the fates themselves; and that memory is carried by one slow promoter, a coarse proxy for
522 chromatin-based inheritance whose single-exponential, symmetric form is a simplification of a process that can be

multi-timescale and bistable. The clinical case studies use literature values for cell-cycle times, resistant fractions and pharmacokinetics rather than patient-specific measurements, so their conclusions concern mechanisms and rankings, not absolute survival times. In the glioblastoma study MGMT is consumed stoichiometrically, one molecule per lesion repaired, so its depletion follows the cumulative exposure as a suicide enzyme’s should. Two things about that model should be read with care. Lesions are not a state variable: they are assumed to form in proportion to dose and to be repaired immediately, so “one molecule per lesion” enters only through the consumption rate and no unrepaired damage accumulates. And the rate constant of that consumption was set so that the initial consumption rate at the standard bolus peak matches the first-order parameterisation it replaces, which makes the two mechanisms comparable but leaves the constant itself unmeasured; bulk depletion time-courses in blood cells and in tumour tissue exist^{54–56} and bound it loosely, and that is why we name the per-lesion constant as the quantity that separates the regimens. MGMT is in any case only one determinant of a temozolomide response that is also mismatch-repair-dependent, and the 1% and 30% expressing fractions are illustrative values that bracket low and high expression rather than measured quantities: promoter methylation shifts expression as a graded quantity across cells rather than switching a discrete subpopulation on, and the dose-dense prediction rests entirely on this two-state idealisation. The memory of the resistant state is expressed in these tumour-scale studies as a number of net population doublings, not of cell divisions, and so is not the division-counted memory of memory-sequencing experiments. A bridge to the Catalyst.jl ecosystem is not part of this release. The benchmark results should be read as demonstrations of what the framework makes possible rather than as verdicts on the methods compared, which were run with default settings on small networks.

Spatial coupling through neighbourhood-dependent rates, cell-cell communication, chromatin states with their own inheritance rules and neural simulation-based inference are natural extensions of the existing forward model; the spatial half of that list is the province of agent-based tumour simulators such as PhysiCell and PhysiBoSS^{31,32}, which model tissue mechanics and microenvironment but not stochastic transcription in dividing cells, so the two are complementary rather than competing. Because the framework is deterministic given a seed, fast enough to simulate tens of thousands of cells in minutes, and writes AnnData files, it can be dropped into single-cell analysis pipelines as a source of mechanistically grounded synthetic data, and into virtual-cell training and evaluation as a source of causal, out-of-distribution contexts.

Methods

Reaction model and propensities

A model is a set of species and reaction channels. Species are molecules (partitioned at division) or promoter states (inherited). For a reaction with reactant counts x_i , stoichiometric coefficients ν_i , rate constant k and volume exponent e , the mass-action propensity is

$$a = k V^e \prod_i \binom{x_i}{\nu_i},$$

where by default $e = 1 - \sum_i \nu_i$, so that zero-order production scales with volume, first-order reactions are volume independent and bimolecular reactions scale as $1/V$; this keeps concentrations invariant under growth. Regulated reactions multiply this by $b + (1 - b)F(\mathbf{c})$, where b is a basal fraction and F combines Hill terms of the regulator concentrations $c_j = x_j/V$, $h_j = c_j^{n_j}/(K_j^{n_j} + c_j^{n_j})$ for activators and $1 - h_j$ for inhibitors, either as a product (AND logic) or as $1 - \prod_j (1 - h_j)$ (OR logic). Promoter switching is represented by explicit promoter species (for instance $G_{\text{off}} \rightleftharpoons G_{\text{on}}$) grouped into promoter groups. Custom propensities can be supplied as functions of counts,

561 volume, parameters and time.

562 Stochastic simulation kernels

563 Within one update step the cell volume and gene copy number are held fixed and the reaction network is advanced
 564 by one of four kernels: the direct method⁵⁷ with propensities updated through a species-reaction dependency
 565 graph; fixed-step Poisson tau-leaping⁵⁸; a hybrid scheme in which a leap that would drive any species negative is
 566 discarded and that interval is simulated exactly instead (the “switching” scheme of version 1, with the discarded
 567 interval resolved exactly rather than clipped); and adaptive tau-leaping with the step-size selection of⁵⁹ and
 568 the critical-reaction partition of⁶⁰, falling back to a fixed number of exact events when the proposed leap is
 569 shorter than $10/a_0$. The hybrid scheme is approximate rather than exact: a leap is kept only when it leaves
 570 every species non-negative, so an accepted step follows the Poisson leap truncated to the non-negative region,
 571 and only the discarded intervals follow the exact process. The discrepancy falls with τ and is bounded by the
 572 Kolmogorov-Smirnov test of Fig. 2d, which the scheme passes at $\tau = 0.05$ and fails at $\tau = 0.2$ for the models
 573 tested; the direct method is used wherever exactness is required (Supplementary Table 1 lists the kernel correctness
 574 tests). Every kernel takes an explicit random number generator.

575 Cells, growth, replication and division

576 Each cell carries its volume V , age, generation, lineage identifiers, gene copy state, growth rate, molecule counts,
 577 a cell-specific parameter vector and its own random number generator (seeded from the parent’s generator, which
 578 makes results independent of the number of threads). Volume grows exponentially, $\dot{V} = \lambda V$, with an optional
 579 log-normal spread of λ between cells. Division is triggered by a sizer ($V \geq V_{\text{div}}$), an adder ($V - V_{\text{birth}} \geq \Delta$) or
 580 a timer ($\text{age} \geq T$), each with a per-cell noise factor drawn at birth. At division the first daughter inherits a
 581 volume fraction $f \sim \mathcal{N}(0.5, \sigma^2)$ (clipped to $[0.05, 0.95]$) and each molecule is assigned to it independently with
 582 probability f (binomial partitioning); a beta-binomial option models over-dispersed partitioning. Promoter groups
 583 are inherited by both daughters when unreplicated and split one copy per daughter, in random assignment, when
 584 replicated. Gene replication, if enabled, doubles the promoter counts (and the propensity of reactions flagged as
 585 copy-number dependent) once the cell has completed a set fraction of its cycle, and is reset at division. Populations
 586 can be started from per-founder states and volumes (for instance a snapshot of a burnt-in constant-size population).
 587 Population control is either constant size (every second daughter replaces a uniformly chosen cell, the behaviour
 588 of version 1), free growth (all daughters kept, with uniform subsampling above a cap and book-keeping of the true
 589 size) or logistic (free growth with an additional hazard $\lambda N/K$).

590 Drug and perturbation layer

591 A perturbation combines a dose schedule $d(t)$ (constant, pulsed, piecewise, one-compartment bolus pharmacoki-
 592 netics or an arbitrary function) with effects evaluated for every cell at every step: a death hazard

$$h = h_{\max} \frac{d^m}{\text{EC}_{50}^m + d^m} \cdot \frac{K^q}{K^q + c_P^q},$$

593 where c_P is the concentration of a protective species (survival probability over a step is $e^{-h \Delta t}$); growth inhibition
 594 by $1/(1 + (d/\text{IC}_{50})^m)$; a dose-independent fitness cost of a cell state, $1 - c_{\max} c_P^q/(K^q + c_P^q)$, on the growth rate;
 595 multiplicative modulation of any parameter by a function of the dose; and genetic perturbations (knockdown,
 596 knockout, over-expression) of any parameter in a random fraction of cells, the flag being inherited by daughters.

Two further effects couple the drug to the cell cycle and to the protein that resists it. A cell-cycle-dependent hazard multiplies h by

$$\beta + (1 - \beta) \exp \left[-\frac{1}{2} \left(\frac{\varphi - \varphi_0}{w} \right)^2 \right],$$

where φ is the cell's progress through its division cycle (0 at birth, 1 at division), so that killing peaks at φ_0 and falls to a fraction β of the peak elsewhere; $\beta = 1$ is the cycle-independent hazard and $\beta = 0$ confines killing to a window of width w around φ_0 . This represents an agent whose lesions are converted into death during replication. Stoichiometric consumption of a protective protein removes molecules of it at rate

$$k dV \frac{c_P}{c_P + K_m},$$

by a Poisson draw over the step. Because lesions form in proportion to the dose, the rate is set by lesion formation alone while the protein is abundant, so the pool follows the cumulative exposure rather than the peak concentration, and falls with what remains once the pool runs low; resynthesis is whatever the reaction model provides. This is the behaviour of a suicide enzyme such as MGMT, which is inactivated by the very lesion it repairs.

Lineage statistics

The lineage table records, for every cell, its parent, founder (clone), generation, birth and end times, fate (divided, died, removed, alive) and its volume and state at birth and at the end of its life. From it we compute Pearson correlations of expression at division between mothers and daughters, sisters and cousins; the lineage autocorrelation r_g between a cell and its ancestor g generations earlier, and the memory timescale τ from $r_g = e^{-g/\tau}$; single-lineage traces by following the ancestral line of a surviving cell; the coefficient of variation squared of concentration across the population versus along single lineages; Luria-Delbrück fluctuation tests (variance of a phenotype across clones grown from single cells compared with the binomial expectation); and MemorySeq-style clonal scores⁶¹, the variance of clone-mean expression divided by its permutation-null expectation.

Observation models

True counts are converted to observations by a scRNA-seq model in which each cell captures every molecule independently with a Beta-distributed efficiency (mean and coefficient of variation set by the protocol), optional multinomial down-sampling to a Poisson-distributed depth, optional additional dropout and batch scaling; an smFISH model with a fixed detection efficiency; and a time-lapse model that samples a species along lineages at a fixed interval with additive or multiplicative measurement noise. Because counts scale with volume, library size correlates with cell size, reproducing the transcriptome-size confound of real data. Outputs are written as CSV, as AnnData-compatible HDF5 files, or as Newick lineage trees.

Inference

Snapshot counts of a gene in non-dividing cells follow the Beta-Poisson law of the telegraph model³⁴, which we evaluate through Kummer's confluent hypergeometric function with rescaled series summation, and maximise by a grid-initialised Nelder-Mead search in log-parameter space. For any model, including populations, we implement sequential Monte Carlo approximate Bayesian computation⁶² with an adaptive tolerance schedule (the α quantile of the previous generation's distances), Gaussian perturbation kernels of variance twice the weighted population variance, importance weights against the prior, and parallel evaluation of the user-supplied distance function; for

count distributions we use the Wasserstein-1 distance between simulated and observed samples⁴³.

Calibration to laboratory data and validation strategy

Reference values were transcribed from the text and tables of the cited articles into `paper/data/` with their provenance (Supplementary Table 4). For the cisplatin case study the model is a resistance gene with slow promoter switching whose protein product protects against a dose-dependent death hazard and against a dose-dependent growth arrest (Methods above), with a 24-h cell cycle (at the fast end of reported U2OS doubling times and consistent with the intermitotic times tracked by⁶). Its six free parameters (activation and inactivation rates, maximal death rate, EC_{50} and Hill coefficient of death, IC_{50} of growth arrest) were fitted by the method of simulated moments: the root-mean-square difference between simulated and observed fractions of cells that died, divided or survived without dividing during 72 h of drug, for the training concentrations (7 and 13 μ M), plus a weak penalty keeping the memory of the resistant state near the two to three generations inferred from lineage correlations, was minimised with the bounded optimiser described below using common random numbers (the same seeds for every candidate). Validation used only quantities not fitted: the fate fractions at the held-out concentration (10 μ M), the biphasic kill curve and its plateau, the extent of lineage correlations in fate (sisters, first, second and third cousins) and the invariance of single-cell division and death times across concentrations. The protocol is deliberately generic: any observable computable from a `PopulationResult` (kill curves, fate fractions, lineage correlations, snapshot distributions) can enter the distance, and the split between training and validation conditions is declared before fitting.

Clinical regimens, treatment outcomes and schedule optimisation

Regimens are built from the schedule primitives of the perturbation layer: `PulsedDose` with a finite number of cycles, `PiecewiseDose` for arbitrary calendars (the S1320 schedule), `daily_boluses` and `cycle_days` for daily oral doses with one-compartment pharmacokinetics, and `AdaptiveDose`, a feedback schedule that withdraws the drug when the population falls below a fraction of its initial size and resumes it when the population regrows above another fraction, evaluated with the current population size at every step⁴⁸. Outcomes computed from a simulation are the net growth rate, the depth of response ($\log_{10}$ of the ratio of the initial to the smallest population), the time to progression (first time the population exceeds a threshold, 1.2, times its running minimum, the analogue of RECIST progression from the nadir; censored at the end of the simulation), the extinction probability over replicates and the cumulative dose actually applied. `optimize_schedule` minimises any such outcome over a box of schedule parameters: a log-spaced grid (or a Latin hypercube when the grid is large) is evaluated with the same random seeds for every candidate, and the best point is refined by Nelder-Mead in a bounded transform of the box; constraints such as a fixed cumulative dose enter as penalties or as the parameterisation (for instance a per-day dose equal to the budget divided by the number of dosing days). The case studies are run at the scale of a tumour rather than a culture: the cell cycle of the model is the net doubling time of the tumour (4 weeks for melanoma, 40 days for glioblastoma). The glioblastoma value lies within the volume doubling times measured for untreated tumours, a median of 21 days in one magnetic-resonance series and 50 days in another^{63,64}; the melanoma value is an assumption, since net doubling times of treated cutaneous melanoma are not established, and it sets the time axis of the melanoma case study rather than the ranking of schedules, which follows from the resistance mechanism. Growth is exponential throughout, which is a simplification: the same measurements favour Gompertzian slowing in large glioblastomas⁶⁴, so absolute regrowth times in Fig. 8 are shorter than a saturating model would give. Gene-expression rates are scaled to the hour (mRNA half-life 7 h and protein half-life 35 h, generic mammalian values; at the tumour time scale both species turn over many times per generation, so the results are insensitive to them and all heritable memory resides in the promoter), and founders are drawn from the

stationary distribution of promoter states. The melanoma case uses a pre-resistant fraction of 1:200 (a per-marker frequency² used as a proxy for the functional pre-resistant fraction, which is not the same quantity), with a memory of five net doublings (the model's generation is the net doubling of the tumour, a modelling choice rather than the cell-division-counted memory of memory sequencing⁶¹) and drug effects at the fractional-killing dose: sensitive cells arrest and die with a half-life of two weeks, cells in the resistant state are protected, and the drug stabilises the resistant state ten-fold (the drug-induced reprogramming of²). Three variants of the resistant state are compared: full protection without cost; a 50% fitness cost in the absence of drug⁴⁹; and partial protection, in which resistant cells grow at half speed under drug. Progression is scored in two ways: from the nadir, as in RECIST (a 20% increase in diameter, taken as a 1.73-fold increase in cell number, after the 8-week lead-in; a well-mixed cell count cannot represent RECIST's absolute 5 mm threshold, its new-lesion and non-target criteria or its sum-of-diameters basis, and it assumes cell number proportional to volume, so this is an analogue rather than the clinical end point), and as loss of control, the time after randomisation at which the population exceeds 120% of its pre-treatment size, the end point of adaptive-therapy studies, which is the one used to compare schedules that deliberately let the tumour regrow. The temozolomide case uses a one-compartment elimination half-life of 2.1 h⁵² (a plasma value from a three-compartment study; the brain-tumour compartment sees roughly a fifth of the plasma exposure with slower kinetics, so this is a surrogate), the regimens of RTOG 0525⁵¹ (200 mg/m² on days 1-5 or 100 mg/m² on days 1-21 of 28-day cycles, six cycles), MGMT-expressing fractions of 1% and 30% for methylated and unmethylated promoters (illustrative values bracketing low and high expression rather than measured fractions; methylation shifts graded expression rather than a bimodal positive fraction), and MGMT either stable or consumed stoichiometrically by the drug, one molecule per lesion repaired ($k = 300$, $K_m = 150$ in the expression above). The rate constant was chosen so that the consumption rate at the standard bolus peak equals that of the dose-scaled first-order degradation used in an earlier version of this model, which isolates the consequence of letting depletion follow the cumulative exposure instead of the peak concentration; this is the suicide-enzyme behaviour that motivated dose-dense schedules.

Random regulatory networks

Random networks are built with a chosen number of activating and inhibiting edges under Erdős-Rényi or preferential-attachment (scale-free) sampling; regulation acts on the transcription rate of constitutive genes or on the promoter activation rate of telegraph genes through AND-combined Hill functions; per-gene rate constants can be drawn log-uniformly from ranges.

Validation protocols

Exactness was tested by Kolmogorov-Smirnov distances between simulated stationary distributions and the analytic laws (Poisson for birth-death, Beta-Poisson for the telegraph model, negative binomial for bursty protein production^{65,66}), by agreement between kernels, and by a two-sample comparison against JumpProcesses.jl⁶⁷ on the same model.

The population and lineage layers were then validated against exact solutions for growing, dividing populations, which exist for this class of model and take different forms for a single lineage and for a population snapshot^{30,35-37}. Four analytically solvable members of that class were simulated in both modes: constitutive production, production in bursts of fixed size, and a two-state promoter, each with a memoryless interdivision time; and volume-scaled synthesis from a gene that replicates at mid-cycle, with a deterministic cycle and exponential volume growth. Single lineages were generated with a mother-machine population control that keeps one daughter at random at every division, so that the recorded cells are independent; population snapshots were generated by free branching

growth with uniform subsampling, which preserves the snapshot distribution. For the memoryless models the exact stationary law of the scheme the simulator runs is the leading eigenvector of one update step, exact kinetics over dt followed by division with probability $1 - e^{-dt/T}$, with the division operator carrying a factor of two in the population mode; it was computed by matrix exponentiation and checked against the closed-form factorial moments, and the step size was scanned to show the discretisation of the interdivision time vanishing. For the replication model the count distribution at every cycle phase is exactly Poisson, and the cycle-phase weights of the population were obtained by iterating the age map of the scheme from the founder ages used, so that no assumption about convergence to a stable age distribution enters. Each simulated distribution was compared with the exact law of its own mode and with that of the other mode, the second as a check that the comparison can tell them apart (Supplementary Note 14). Population-level properties were compared with the analytic expectations for cell-size scaling and concentration homeostasis²⁷, partitioning noise⁶⁸, lineage versus population statistics¹¹ and cell-cycle-dependent burst parameters¹⁰. Persister simulations were compared with the quantitative signatures reported by^{6, 42} and⁹, and with the phenotypic-selection model of⁴¹. All scripts, seeds and outputs are in the `paper/` directory of the repository.

Identifiability and numerical robustness of the calibration

Because six parameters were fitted to six measured fractions, the calibration was examined in three ways. A conditional profile moves one parameter at a time over its full range on a log grid of eleven points, holding the others at their fitted values, and records the root-mean-square error of the fate fractions without the memory prior; parameters whose error stays within the binomial standard error of the measurement over a wide range are not determined by these data. A two-dimensional slice does the same over the memory of the resistant state and the fraction of cells in it, the two quantities that trade off most directly, with the death parameters held at their fitted values; a slice was used in preference to a profile with re-optimisation because, at this cost per evaluation, the inner search did not reliably recover the fitted solution and reported spuriously poor fits at short memory. Numerical robustness was checked by halving the integration step and comparing the simulated fractions with the seed-to-seed spread at the fitted parameters.

Because the lethality of cisplatin is enhanced during replication, the position of a cell in its division cycle when the drug arrives is a competing explanation for the dose-invariance of death timing and for the fraction of cells that survive without dividing. To test it, the same six parameters were refitted together with a seventh, the cycle-independent fraction β of the death hazard, with the peak of killing at the middle of the cycle ($\varphi_0 = 0.5$, $w = 0.15$), under the same objective, bounds, optimiser budget and seeds as the main fit. The two fits are compared by the root-mean-square error of the fate fractions without the memory prior, by their predictions at the held-out concentration, and by a conditional profile over β with the remaining parameters held at the cycle-dependent fit. The same gate ($\beta = 0.25$, $\varphi_0 = 0.5$, $w = 0.15$) was then applied to the persister simulations of Fig. 4 and the melanoma simulations of Fig. 8a-e, everything else unchanged, to test whether their conclusions depend on the hazard being cycle-blind (Supplementary Note 13). The window position is a modelling choice rather than a claim about either agent: what is tested is robustness to a strong cell-cycle gate of any kind.

Statistics and reproducibility

No statistical method was used to predetermine sample size, and no experimental data were generated in this study; sample sizes are the numbers of simulated cells, founders or seeds stated in each figure legend and in Supplementary Table 2. No data were excluded. Randomisation and blinding do not apply: every result is a deterministic function of an explicit random seed, and all seeds are fixed in the scripts that generate the

756 figures. Replicates are independent simulations differing only in seed; where a mean and a standard deviation
757 are reported, the number of seeds is given in the legend. Kolmogorov-Smirnov distances are compared with the
758 99% critical value for the stated sample size. Distributional agreement between kernels and with an independent
759 implementation was assessed by two-sample Kolmogorov-Smirnov statistics. Correlations are Pearson correlations
760 unless stated otherwise; correlations of binary fates between related cells are phi coefficients. Posterior summaries
761 are medians, because the posteriors are asymmetric on the log scale. Multithreaded runs give results identical to
762 single-threaded runs, because each cell carries its own random number generator seeded from its parent.

763 Use of artificial intelligence

764 The software and the first draft of this Article were produced with Claude (Anthropic), a large language model,
765 under the direction and supervision of the author. Specifically, Claude wrote the Julia source of the package, the
766 simulation and plotting scripts, the test suite and the first draft of the text, working from the author's specification
767 of the scope, the models, the validation strategy and the case studies. The author defined every scientific question
768 addressed here, chose the models, parameters and reference data, independently verified all numerical results
769 against the generated output tables, checked every claim against its cited source, and wrote the final version of
770 the manuscript; the author takes full responsibility for the content. The model is not an author, in keeping with
771 journal policy on generative artificial intelligence.

772 No display item in this Article is generated by a generative model. Every figure and table is produced deter-
773 ministically from simulation output by the scripts in the `paper/` directory: the panels by `matplotlib` from the
774 comma-separated tables in `paper/output/`, and every number quoted in the Results and the Supplementary
775 Information by `paper/manuscript/fill_numbers.py`, which substitutes values read from those tables into the
776 manuscript source. Figure 1 is a schematic drawn by `paper/fig1_schematic.py`. Re-running the scripts from
777 the fixed seeds reproduces all of them.

778 Data availability

779 The laboratory and clinical reference values used for calibration and validation were transcribed from the published
780 articles cited in Supplementary Table 4 and are provided, with their provenance, in the `paper/data/` directory
781 of the software repository (<https://github.com/ayoublasri/Biomodelling.jl>). All other data in this Article are
782 simulated and are regenerated from fixed seeds by the scripts in the `paper/` directory; the generated tables are
783 archived with the software release.

784 Code availability

785 Biomodelling.jl 2.0 is open source under the MIT licence at <https://github.com/ayoublasri/Biomodelling.jl>, with
786 documentation, tutorials and a test suite that runs under continuous integration. Every figure is regenerated
787 from fixed seeds by the scripts in `paper/`. After instantiating the environment (`julia --project=paper -e`
788 `'using Pkg; Pkg.develop(path="."); Pkg.instantiate()'`), the simulation data are produced by `julia`
789 `--project=paper -t 4 paper/run_all.jl`, the Python network-inference and imputation benchmarks of Fig.
790 5b–d by `python paper/scripts/benchmarks_python.py`, and the identifiability analysis of Supplementary Fig.
791 3 by `julia --project=paper paper/scripts/fig7_calibration.jl profile`; the panels are then assembled
792 by `python paper/fig1_schematic.py`, `python paper/plot_all.py` and `python paper/plot_supplement.py`.
793 Full instructions, including the four-thread timings of Supplementary Table 3, are in `paper/README.md`.

Software

Biomodelling.jl 2.0 is written in Julia (≥ 1.10) and depends on Distributions.jl, PoissonRandom.jl, QuadGK.jl, Optim.jl, SpecialFunctions.jl and StatsBase.jl; HDF5.jl is an optional extension. Benchmarks used absolute Pearson and Spearman correlations and scikit-learn random forests re-implementing GENIE3⁶⁹, MAGIC⁷⁰ and kNN-smoothing⁷¹. The test suite (Aqua quality checks and statistical tests) runs under GitHub Actions on Julia 1.10 and the latest release.

Figure legends

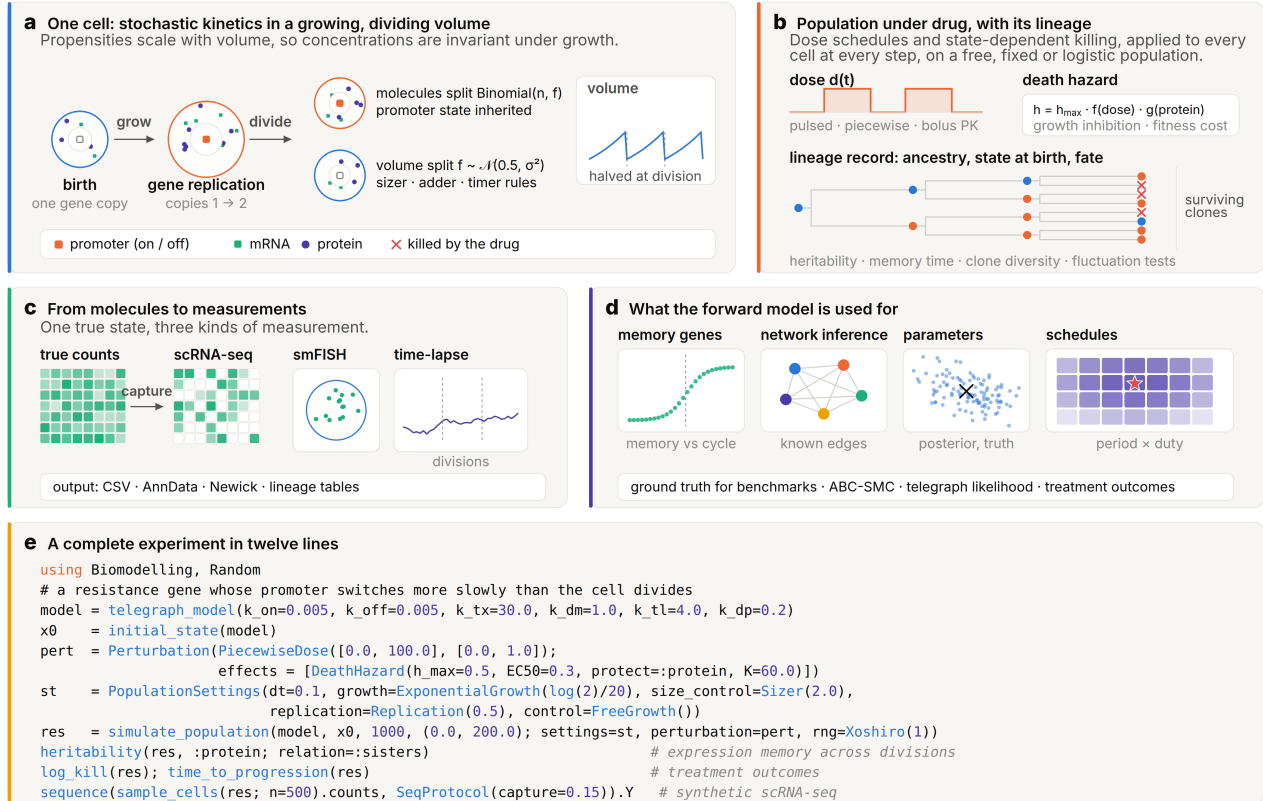
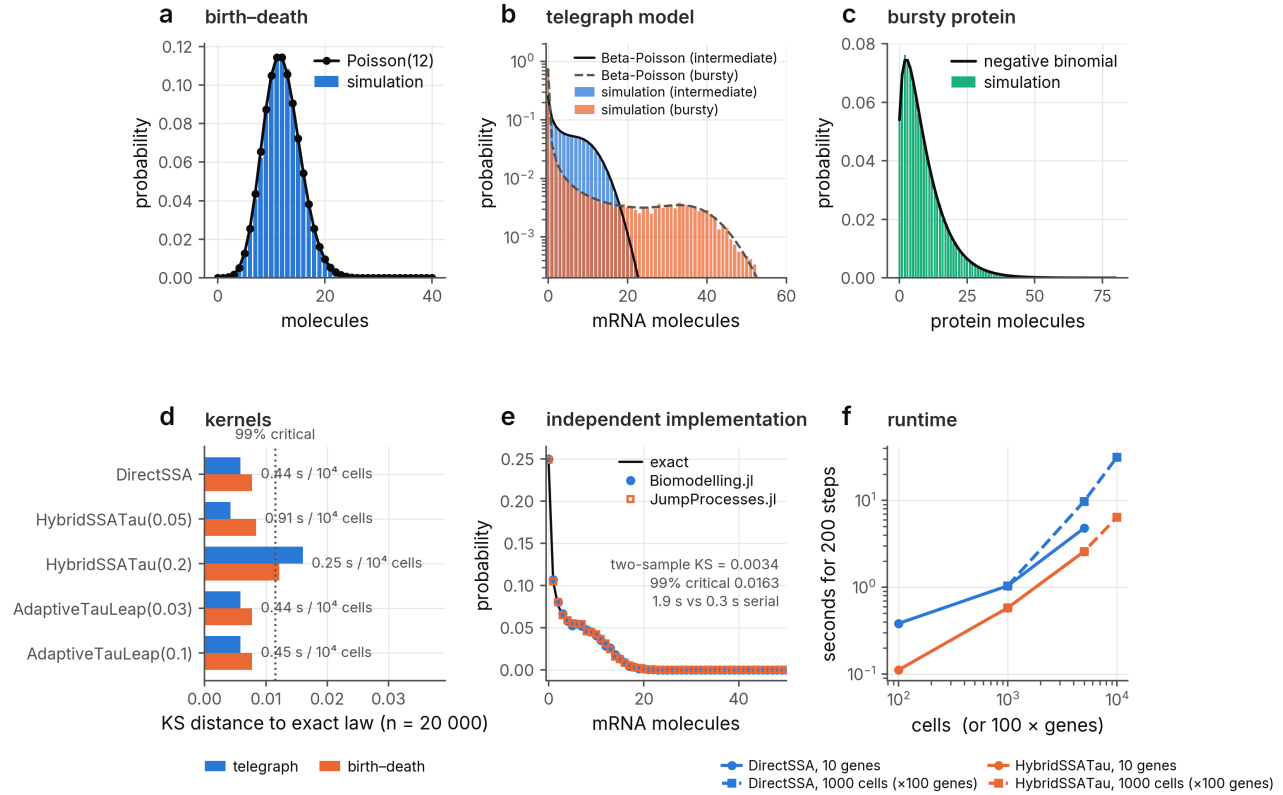
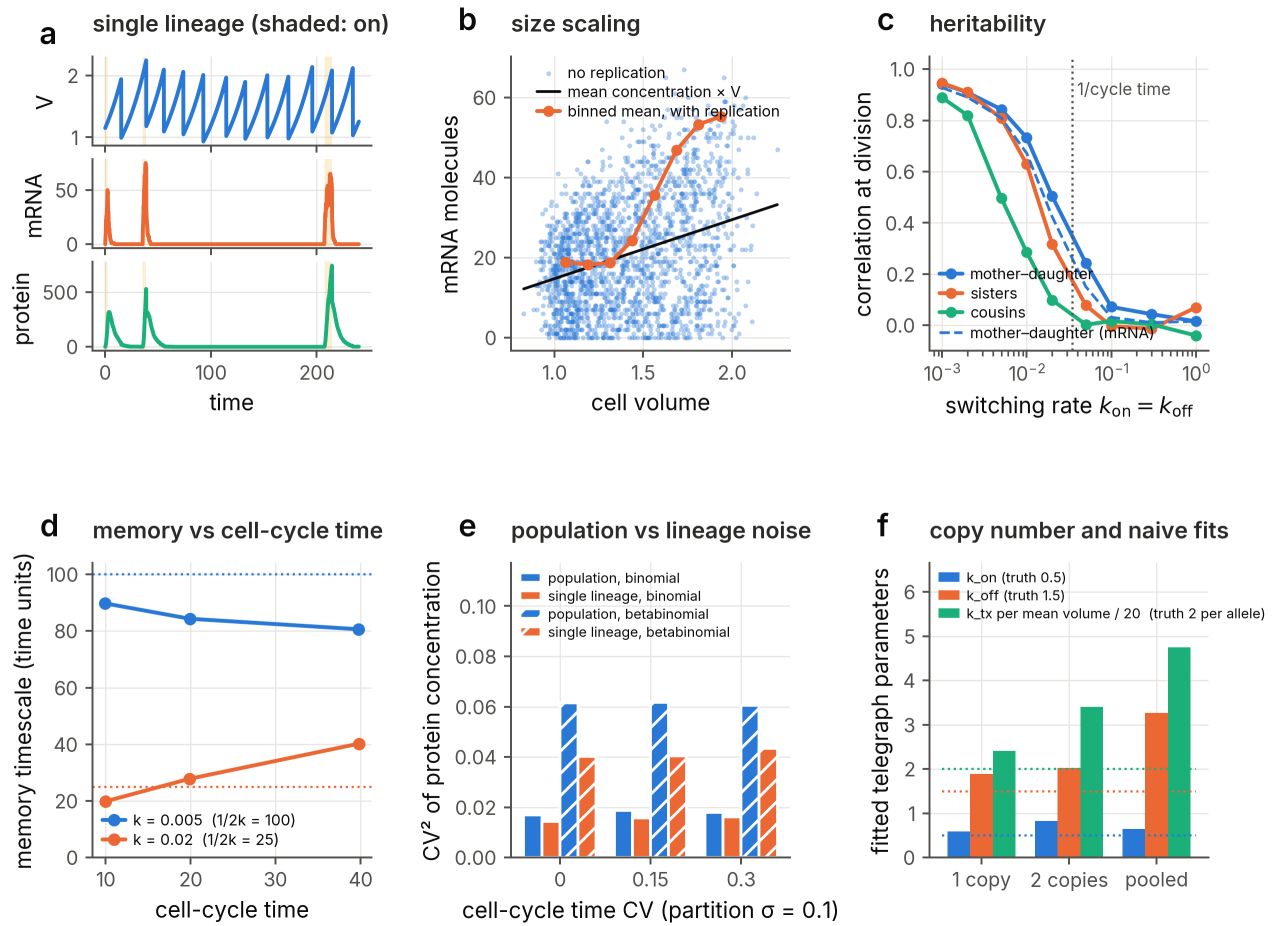


Fig. 1 | The Biomodelling.jl 2.0 framework. **a**, One cell. Reactions run at the cell's current volume, with propensities scaled so that concentrations are invariant under growth; genes replicate at a set point of the cycle; division splits every molecule binomially while both daughters inherit the promoter state. Inset: exponential volume growth, halved at each division. **b**, Populations under free growth, constant-size or logistic control, with dose schedules and a state-dependent death hazard evaluated for every cell at every step, and the lineage record from which heritability, autocorrelation, memory timescales, clone diversity and fluctuation tests are computed; crosses mark cells killed by the drug. **c**, Observation models turn true counts into scRNA-seq counts with capture, depth, dropout and batch effects, into smFISH spots, or into time-lapse reporter traces; outputs are CSV, AnnData and Newick. **d**, Uses of the forward model: detection of memory genes, benchmarking of network and perturbation inference against known ground truth, likelihood-free parameter inference, and schedule optimisation over the period and the fraction of time on drug (star, optimum). **e**, A complete experiment in twelve lines of code.



814

815 **Fig. 2 | The engine matches exact laws and an independent implementation.** **a**, Stationary distribution
816 of a birth-death process (bars, 50,000 simulated cells; line, Poisson law). **b**, Telegraph model in an intermediate
817 and a bursty regime against the Beta-Poisson law. **c**, Bursty protein production against the negative binomial. **d**,
818 Kolmogorov-Smirnov distance to the exact law for every kernel (n = 20,000; dotted line, 99% critical value) with
819 wall-clock time per 10^4 cells. **e**, Telegraph model simulated with Biomodelling.jl and with JumpProcesses.jl on
820 the same model; the two-sample Kolmogorov-Smirnov statistic lies below its 99% critical value. **f**, Runtime of
821 population simulations against the number of cells (10 genes) and genes (1,000 cells) for the exact and hybrid
822 kernels.



823

824 **Fig. 3 | Growth, division and partitioning generate memory, size scaling and cell-cycle effects.** **a**, A
825 single lineage: volume, mRNA and protein of a slowly switching gene, with the promoter-on state shaded. **b**,
826 mRNA count against cell volume for a fast-switching gene without replication (points; line, mean concentration
827 $\times V$) and with gene replication at mid-cycle (binned means). **c**, Correlation at division between mothers
828 and daughters, sisters and cousins as a function of the promoter switching rate (protein concentration, solid;
829 mother-daughter mRNA, dashed); dotted line, inverse cell-cycle time. **d**, Memory timescale against the measured
830 cell-cycle time for two switching rates; dotted lines, $1/(k_{\text{on}} + k_{\text{off}})$. **e**, Squared coefficient of variation of a stable
831 protein's concentration across the population and along single lineages, for increasing partitioning noise and
832 cell-cycle-time noise. **f**, Telegraph parameters fitted to cells before replication (one gene copy), after replication
833 (two copies) and pooled. The transcription rate is plotted per unit cell volume and divided by twenty, so that all
834 three series share an axis and the dotted lines mark the true per-allele values ($k_{\text{on}} = 0.5$, $k_{\text{off}} = 1.5$, and 2 for the
835 scaled transcription rate); the raw fitted transcription rates are in `fig3f_copynumber.csv`.

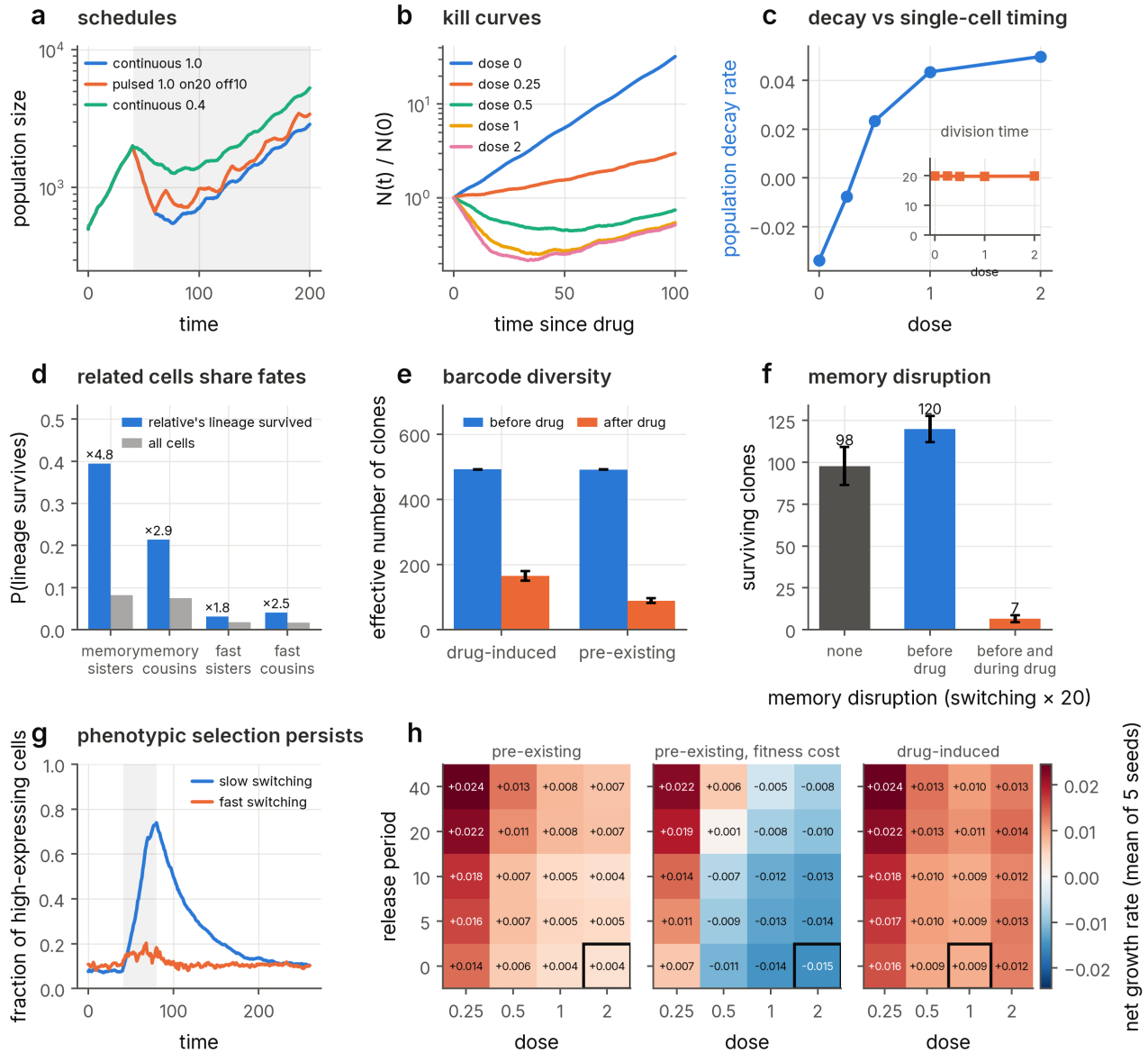


Fig. 4 | A single resistance gene with slow promoter switching reproduces persister biology. **a**, Population size under continuous, pulsed and reduced-dose schedules (shading, treatment window). **b**, Kill curves at increasing dose. **c**, Net population decay rate over the first 30 time units of treatment against dose, with the mean division time of cells born under drug (inset), which is unchanged. **d**, Probability that a cell's lineage survives the treatment (at least one descendant alive at the end of the run) given that its sister's or cousin's lineage survived, against the marginal survival probability, for cells born in the 25 time units before the drug; memory gene and fast-switching control, with the fold enrichment above each bar. **e**, Effective number of clones (barcodes) before and after drug for pre-existing and drug-induced tolerance (mean $\pm$ s.d. of three seeds). **f**, Surviving clones after drug without memory disruption, with promoter switching accelerated 20-fold from two cell-cycle times before the drug until the drug arrives, and with the acceleration maintained during exposure (mean $\pm$ s.d. of five seeds). **g**, Fraction of high-expressing cells before, during and after a drug pulse for slow and fast switching. **h**, Net population growth rate over a 200-time-unit course as a function of release period (time units without drug between 20-time-unit exposures) and dose, for pre-existing tolerance without and with a fitness

cost of the resistant state and for drug-induced tolerance; each cell of the map is the mean of five independent seeds and the box marks the most effective schedule.

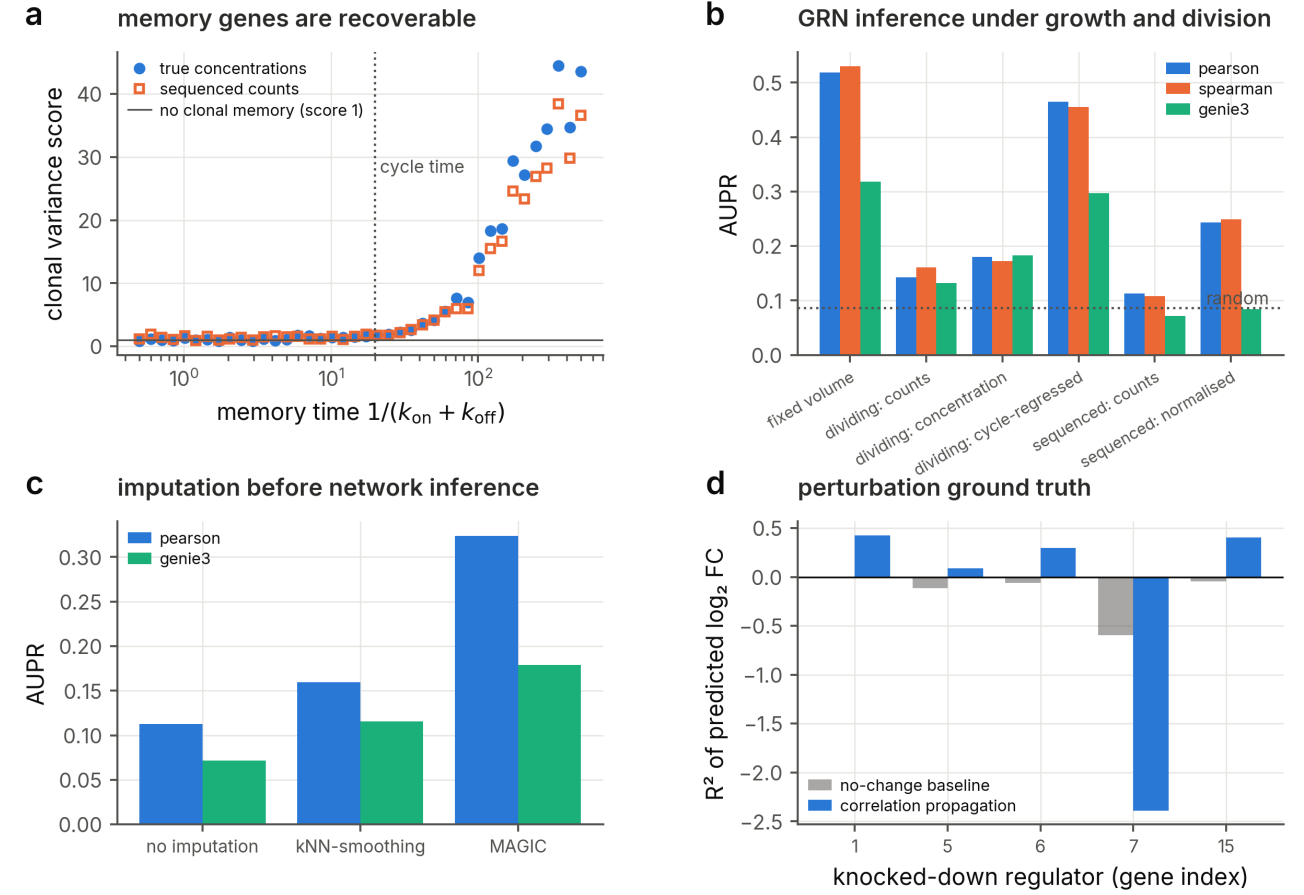
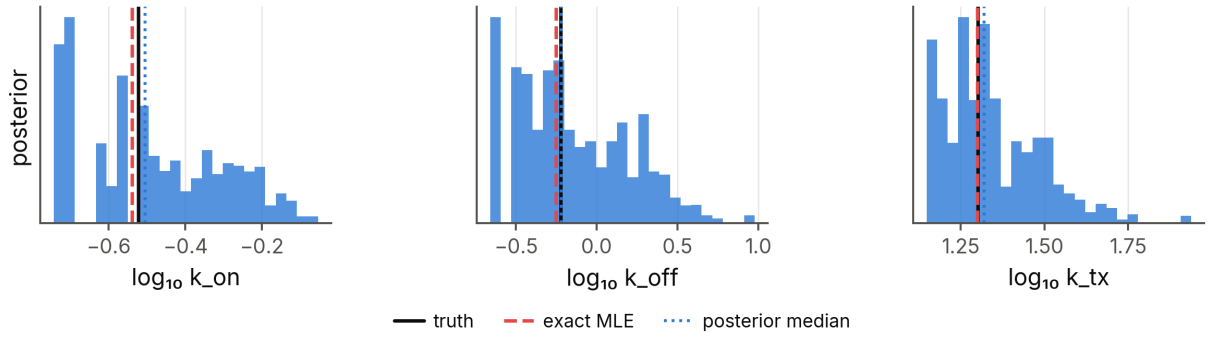
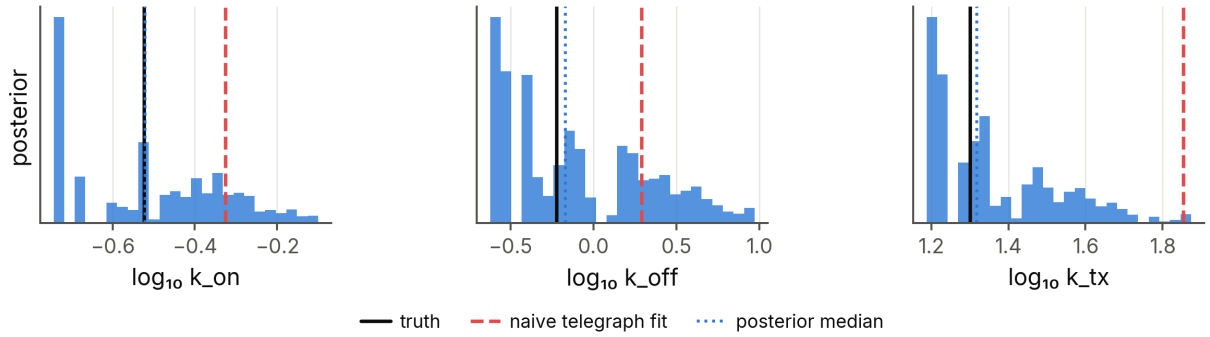


Fig. 5 | Ground-truth benchmarks. **a**, Clonal variance score against the known memory time of 40 independent genes, from true concentrations and from sequenced counts; dotted line, cell-cycle time. **b**, Area under the precision-recall curve of correlation (Pearson, Spearman) and GENIE3 network inference on the same 30-gene network observed at fixed volume, in a dividing population (raw counts, concentrations, and concentrations regressed on the cell-cycle covariates copy number and age) and after sequencing (raw and library-size normalised counts); dotted line, random ranking. **c**, Effect of kNN-smoothing and MAGIC imputation before inference. **d**, Coefficient of determination of predicted $\log_2$ fold changes after knocking down the five most connected regulators, for a no-change baseline and for propagation of observational correlations.

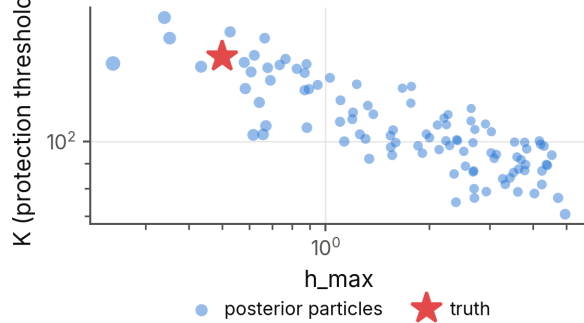
a non-dividing cells: ABC vs exact MLE



b dividing population: division-aware ABC vs naive fit



c drug parameters from kill curve and sister fates



d posterior predictive check

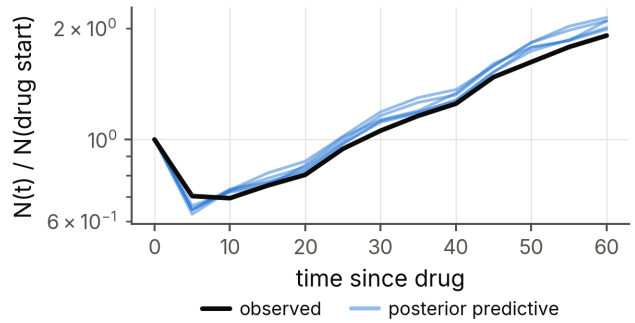


Fig. 6 | Inference with the forward model. **a**, Posterior distributions of telegraph parameters from ABC-SMC on a snapshot of non-dividing cells, with the truth and the exact maximum-likelihood estimate. **b**, The same for a snapshot of a growing and dividing population: the naive telegraph fit is biased while division-aware ABC recovers the truth. **c**, Posterior particles for the maximal death rate and the protection threshold inferred from a kill curve and sister fate concordance: the two parameters trade off along a ridge that contains the truth. **d**, Posterior predictive kill curves simulated at the posterior median against the observed curve.

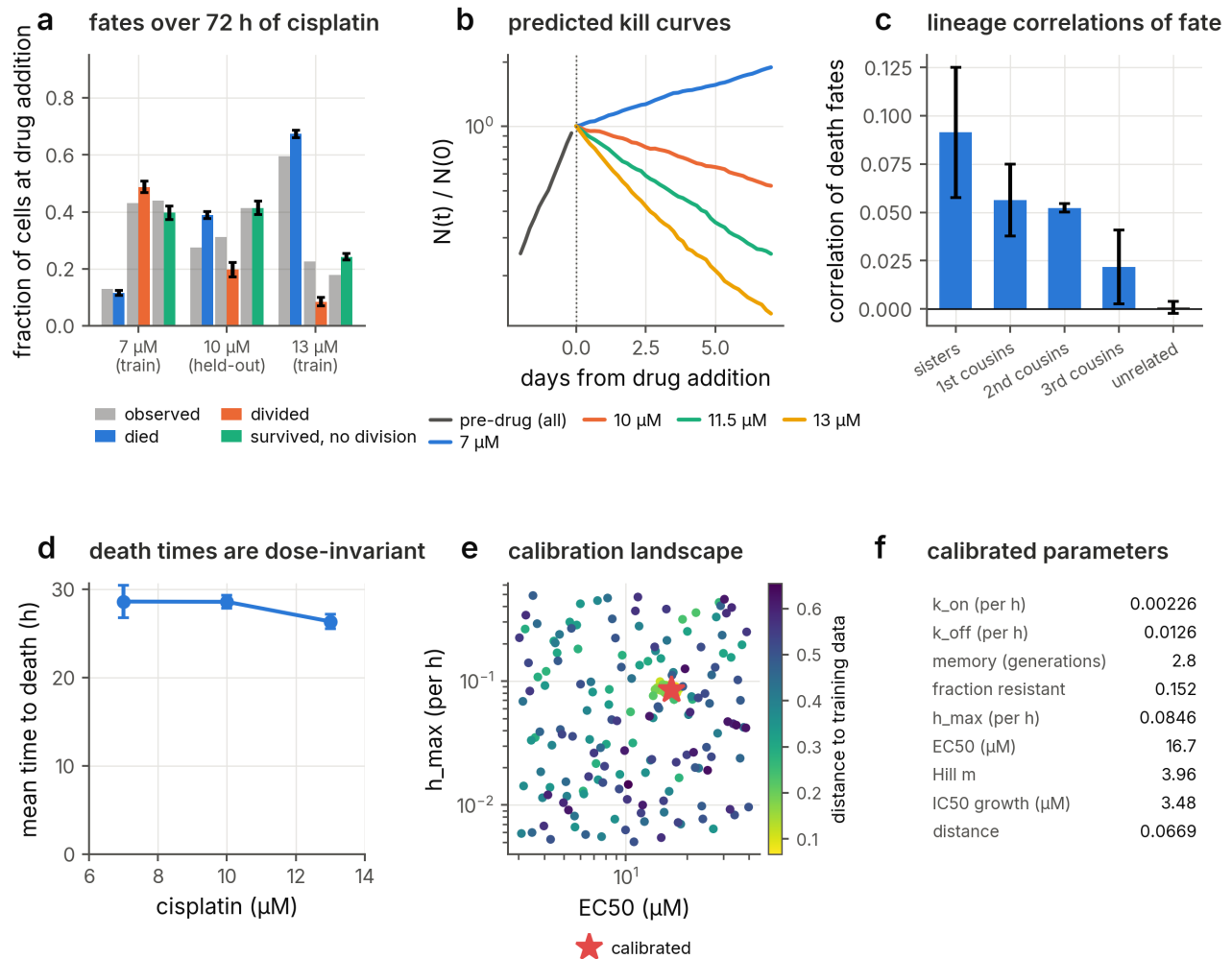
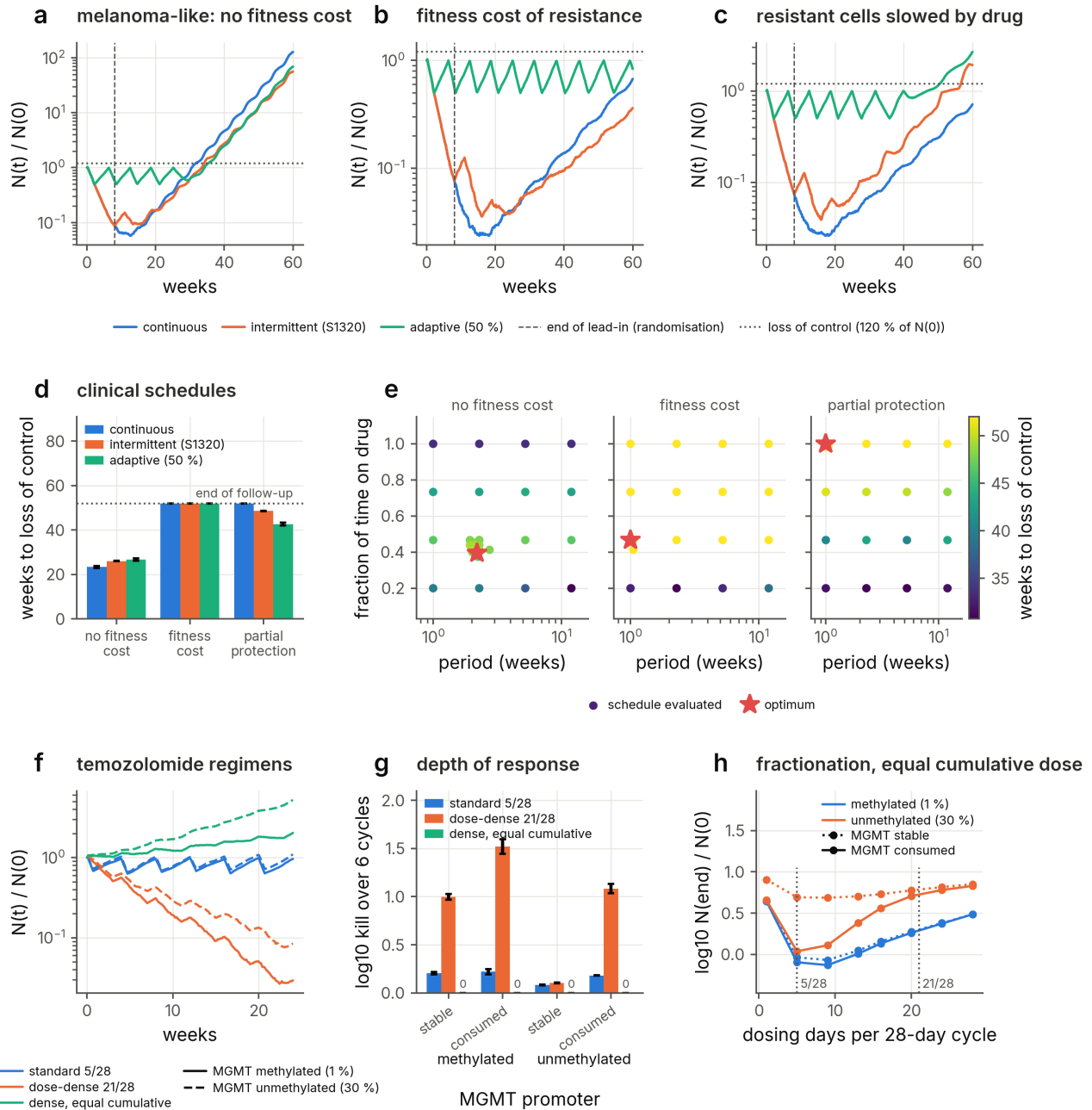


Fig. 7 | Calibration to laboratory time-lapse data and validation on held-out conditions. **a**, Fates over 72 h of cisplatin of U2OS cells present at drug addition, observed by time-lapse microscopy (grey; ref.⁶) and simulated by the calibrated model (colours; mean $\pm$ s.d. of four seeds); the model was calibrated on 7 and 13 μM and the 10 μM condition is predicted. **b**, Predicted population size at the three concentrations and at 11.5 μM , from two days before the drug to seven days of exposure. **c**, Correlation of death fates between sisters, first, second and third cousins present at drug addition, and between unrelated cells (mean $\pm$ s.d. of three seeds), against the observation that correlations extend to second but not third cousins. That observation is reported graphically in ref.⁶ with no numeric coefficients, so the comparison is of pattern rather than of value. **d**, Mean time to death of cells present at drug addition against concentration (mean $\pm$ s.d. of four seeds); division times are not shown because too few cells born under drug divided again to estimate them (at most four per condition). **e**, Calibration landscape: every parameter set evaluated by the optimiser, coloured by its distance to the training data. **f**, Calibrated parameters.



881

882 **Fig. 8 | Dose and schedule optimisation against clinical regimens.** a–c, Population size, relative to its
 883 pre-treatment value, under continuous dosing, the intermittent schedule of the SWOG S1320 trial (8-week lead-in,
 884 dashed line, then 3 weeks off and 5 weeks on) and adaptive therapy (drug withdrawn below 50% of the initial size,
 885 resumed at the initial size) for a melanoma-like population with a rare, heritable pre-resistant state, when the
 886 resistant state carries no fitness cost (a), when resistant cells grow 50% slower in the absence of drug (b), and
 887 when resistant cells are only partly protected and grow at half speed under drug (c); dotted line, 120% of the
 888 initial size. d, Weeks after randomisation until the population exceeds 120% of its pre-treatment size (loss of
 889 control; mean $\pm$ s.d. of four seeds, censored at 52 weeks). e, The same end point over intermittent schedules of
 890 given period and fraction of time on drug after the lead-in, for every schedule evaluated by the optimiser (star,
 891 best), for the three mechanisms. f, Population size under the standard (5 of 28 days) and dose-dense (21 of 28

days) temozolomide regimens of the RTOG 0525 trial and a 21-day regimen at the standard cumulative dose, with one-compartment pharmacokinetics (elimination half-life 2.1 h, the value of the cerebrospinal-fluid population model; the plasma value is 1.8 h) and stoichiometric consumption of MGMT by the drug, one molecule per lesion repaired, for populations in which 1% (methylated promoter, solid) or 30% (unmethylated, dashed) of cells express MGMT. **g**, Depth of response over six cycles for the three regimens, the two populations and MGMT either stable or consumed by the drug (mean $\pm$ s.d. of two seeds); a bar at zero marks a regimen that achieves no net kill (the 21-day schedule at the standard cumulative dose). **h**, Final population size against the number of dosing days per 28-day cycle at the standard cumulative dose.

Acknowledgements

Version 1 of the framework was developed with Marc Sturrock and Vahid Shahrezaei, whose contributions to that work are described in ref.³³. This Article and version 2.0 of the software were authored with Claude (Anthropic), a large language model, under the direction and supervision of the author; what the model did and what the author verified independently are set out in the Methods, under Use of artificial intelligence.

Author contributions

A.L. conceived the study, defined its scope, the validation strategy and the case studies, supervised the work, verified all numerical results and biological interpretations against their sources, and wrote the final version of the manuscript. The implementation of the software, the simulations, the analyses, the figures and the first draft of the manuscript were produced with Claude (Anthropic) under the author's direction, as described in the Methods. Claude is not an author and no display item was produced by a generative model.

Competing interests

The author declares no competing interests.

Additional information

Supplementary information accompanies this Article. Correspondence and requests for materials should be addressed to A.L.

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